SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Design and Control of an Aerodynamic Surface-Enhanced Multirotor

William Constantin Rosenhahn

SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Design and Control of an Aerodynamic Surface-Enhanced Multirotor

Author:	William Constantin Rosenhahn
Examiner:	Prof. Dr.-Ing. Markus Ryll
Supervisor:	Lukas Pries
Submission Date:	01.12.2025

I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 01.12.2025 William Constantin Rosenhahn

Acknowledgments

"I would like to thank Pablo Kirby for developing the improved platform variant and supporting the experimental validation, and for the collaborative work environment."

Abstract

Contents

1 Introduction

1.1 Motivation

Micro air vehicles (MAVs) with efficient autonomous navigation can strengthen applications such as search-and-rescue and last-mile delivery, where safety, robustness, and endurance are critical. Conventional quadrotors offer agility and precise control but are power-inefficient in sustained forward flight. Fixed-wing platforms are efficient but cannot hover and are less maneuverable in confined spaces. Hybrid vertical take-off and landing (VTOL) concepts improve mission versatility but increase mechanical and control complexity. This thesis investigates an intermediate design: a multirotor augmented with fixed aerodynamic surfaces to harvest passive lift during horizontal motion while keeping multirotor agility.

1.2 Problem statement and scope

In the presence of aerodynamic surfaces, hard-to-model aerodynamic forces become significant and challenge controller design. We aim to develop a platform and control strategy that:

- preserves quadrotor agility while improving forward-flight efficiency via passive lift,
- tracks agile trajectories accurately without requiring aerodynamic parameter identification, and
- remains robust to modeling errors and disturbances.

The central question is whether an Incremental Nonlinear Dynamic Inversion (INDI)-based controller can achieve accurate trajectory tracking on an aerodynamic surface-enhanced quadrotor without an explicit aerodynamic model.

1.3 Contributions

This thesis presents:

- a design of an aerodynamic surface-enhanced quadrotor in X-wing configuration,
- a dynamics model combining a standard quadrotor 6-DoF model with simplified quadratic lift/drag,
- a trajectory-tracking controller based on geometric control and INDI, including a coordinated-turn option, and
- an experimental evaluation: thrust-map identification, agility/controllability, aerodynamic disturbance characterization, and efficiency assessment.

1.4 Thesis organization

Chapter 2 reviews related work and motivates the chosen design. Chapter 3 derives the model. Chapter 4 details the platform. Chapter 5 presents the controller. Chapters 6 and 7 describe the setup and experiments. Chapter 8 concludes.

2 Related Work

Designing a UAV that combines agility and energy efficiency while remaining controllable across hover and forward flight requires balancing canonical architectures: pure multirotors, fixed-wing aircraft, and hybrid configurations such as tailsitters, tiltrotors, and quadplanes. In addition, several studies investigate aerodynamic augmentation of multicopters through fixed lifting or streamlining surfaces to mitigate the inherent forward-flight inefficiency of conventional quadrotors.

2.1 Baselines: Agility vs. Efficiency

Pure multirotors Quadrotors excel in agility and controllability during hover and low-speed flight. Incremental Nonlinear Dynamic Inversion (INDI) has become a robust inner-loop control approach for such vehicles, proven to handle modeling errors and aerodynamic disturbances without requiring a detailed aerodynamic model [SCM10; Tzo+21]. These controllers enable precise tracking of aggressive trajectories, but the need to tilt thrust for lift makes sustained forward flight energetically inefficient.

Fixed-wing aircraft Fixed-wing UAVs achieve superior cruise efficiency due to aerodynamic lift, yet cannot hover or sustain stationary flight.

Hybrids: tailsitters and tiltrotor/tilt-wing systems Hybrid VTOL designs seek to merge both regimes. Tailsitter configurations have demonstrated global trajectory-tracking and agile uncoordinated or knife-edge flight using global INDI frameworks [TK22; TK21]. Tilt-wing concepts have been modeled and simulated with detailed transition dynamics, for example by Daud Filho et al. [Da24]. However, these systems remain mechanically and algorithmically complex, with significant coupling effects during transitions.

2.2 Controller Classes Across the Spectrum

Geometric and differential-flatness-based control form the theoretical foundation for aggressive UAV tracking. Goodarzi et al. introduced a geometric adaptive variant capable of handling parameter uncertainty on SE(3) [GLL15], while Mellinger and Kumar presented a polynomial trajectory generation and tracking approach based on differential flatness [MK11]. INDI, originally formulated for robust flight control in fixed-wing aircraft [SCM10; Kam+18], has been successfully applied to agile multirotors as a disturbance-rejection strategy [Tzo+21]. These works collectively show that modern control architectures can achieve strong tracking and robustness without precise aerodynamic modeling.

2.3 Aerodynamic Augmentation on Multicopters

A separate line of research attaches fixed lifting or streamlining surfaces to quadrotors to improve cruise efficiency while retaining standard control structures.

Wings Dawkins and DeVries integrated small wings on a micro-quadrotor and quantified a clear trade-off: roughly 35% lower power consumption in forward flight, at the cost of about 45% more hover power due to added mass and drag. The wing configuration became advantageous above approximately 3–5 m/s, depending on angle of attack and rotor interaction [DD18]. These findings indicate that properly sized wings can yield meaningful efficiency gains with minimal impact on control authority.

Xiao et al. attached a fixed lifting wing to a 1.2 kg multirotor and reported 50.14% less electrical power at 15 m/s relative to the bare quad, without requiring controller changes [XQa20]. This demonstrates that aerodynamic surfaces can roughly halve the energy per distance at moderate cruise speeds while maintaining standard quadrotor control.

Shrouded and ducted designs Research on shrouded rotors likewise shows aerodynamic performance benefits and noise reduction potential for compact UAVs [HBC14]. Such configurations further highlight the design space between pure multirotors and fully morphing hybrids.

2.4 Quantitative Comparison and Trade-offs

Agility INDI-based multirotor controllers achieve aggressive tracking and strong disturbance tolerance [SCM10; Tzo+21]. While multi-g performance has been shown in other studies, those data are not included here and thus not quantified.

Efficiency Empirical evidence from fixed-surface designs shows substantial forward-flight gains: approximately 35%–50% lower cruise power between 5–15 m/s [DD18; XQa20]. Hover efficiency is modestly degraded due to additional mass and drag.

Controllability and transitions Multirotors and their aerodynamically augmented variants retain full control authority in both hover and forward flight using the same actuators and control allocation, avoiding transition complexity. Hybrid systems such as tilt-wings require coordinated transition planning and mode-dependent control allocation [Da24].

Implementation complexity Adding fixed aerodynamic surfaces is mechanically simple and controller-agnostic. Tilt-wing or tailsitter hybrids, by contrast, involve additional mechanisms and complex transition management [TK22; Da24].

2.5 Gaps and Open Issues

Three key gaps remain evident:

1. Few published experiments report both aggressive maneuverability and detailed cruise-efficiency metrics for fixed-surface quadrotors.
2. Rotor-wake and wing-interaction effects across the speed envelope are incompletely modeled, leaving uncertainty in disturbance behavior.
3. Hybrid-VTOL studies document transition dynamics but rarely quantify lateral-acceleration or coordinated-turn performance under real aerodynamic loads [Da24].

2.6 Why a Quadrotor with Fixed Aerodynamic Surfaces?

The evidence from verified literature supports this design rationale:

1. **Preserve controllability and robustness:** INDI and geometric/differential-flatness controllers provide reliable tracking and disturbance tolerance without detailed aerodynamic modeling [SCM10; GLL15; Tzo+21].
2. **Gain measurable efficiency improvements:** Fixed aerodynamic surfaces yield 35–50% cruise-power reduction in the 5–15 m/s range [DD18; XQa20].
3. **Keep low mechanical complexity:** No tilting actuators or transition control are required; standard multirotor architecture suffices [Da24].

Given the criteria—agility, cruise efficiency, controllability, and simplicity—the literature favors a quadrotor with fixed aerodynamic surfaces over more complex hybrids.

3 Dynamics Model

This chapter presents the six-degree-of-freedom (6-DoF) rigid-body model for the X-wing quadrotor. The formulation uses a standard quadrotor body model augmented with a simple aerodynamic wing model (quadratic in airspeed) and lumps unmodeled effects into disturbance terms. The model is used for simulation, parameter identification, and to document the quantities that the controller deliberately does *not* model explicitly (because INDI compensates them incrementally).

3.1 List of Symbols

Table 3.1: Symbols used in the dynamics model.

Symbol	**Description**	**Unit**
$\mathcal{W}$	World (inertial) frame (ENU)	—
$\mathcal{B}$	Body-fixed frame at CoG	—
$\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$	Body-frame unit vectors	—
$\mathbf{p}$	Position in world frame	m
$\mathbf{v}$	Velocity in world frame	m/s
R	Rotation matrix $\mathcal{B} \to \mathcal{W}$	—
$\boldsymbol{\omega}$	Angular velocity in body frame	rad/s
$\mathbf{g}$	Gravity vector $[0, 0, -g]^\top$	m/s^2
m	Vehicle mass	kg
J	Inertia matrix in body frame	kg·m^2
$\mathbf{F}_g$	Gravitational force	N
f_i	Thrust produced by rotor i	N
$\mathbf{u}$	Control input vector $[f_1, f_2, f_3, f_4]^\top$	N
ω_i	Angular speed of rotor i	rad/s
k_t	Thrust coefficient	N·s^2·rad^{-2}
b_i	Static bias in thrust model for rotor i	N
k_Q	Torque coefficient	N·m·s^2·rad^{-2}

Continued on next page

Table 3.1 – continued from previous page

Symbol	**Description**	**Unit**
Q_i	Reaction torque of rotor i	N·m
c_i	Prop rotation sign (± 1)	—
$\mathbf{v}_w$	Wind velocity in world frame	m/s
$\mathbf{v}_a$	Airspeed in body frame	m/s
V	Airspeed magnitude	m/s
α	Angle of attack	rad
β	Sideslip angle	rad
ρ	Air density	kg/m^3
q	Dynamic pressure $= \frac{1}{2}\rho V^2$	N/m^2
S	Wing reference area	m^2
b	Wing span	m
AR	Aspect ratio b^2/S	—
a_α	Lift-curve slope	rad^{-1}
a_β	Side-force slope (or C_{Y_β})	rad^{-1}
C_L, C_D, C_Y	Lift, drag, side-force coefficients	—
C_{D0}	Zero-lift drag coefficient	—
k	Induced drag factor	—
$\mathbf{F}$	Total force in world frame	N
$\mathbf{F}_T$	Rotor thrust force in world frame	N
$\mathbf{F}_{aero}$	Aerodynamic force in body frame	N
$\boldsymbol{\tau}$	Total moment in body frame	N·m
$\boldsymbol{\tau}_T$	Rotor-generated moment	N·m
$\boldsymbol{\tau}_{aero}$	Aerodynamic moment	N·m
$\mathbf{r}_i$	Arm vector to rotor i	m
$\mathbf{r}_w$	Aerodynamic center offset	m
$\hat{\mathbf{t}}_i$	Rotor i thrust direction unit vector	—
G	Allocation matrix (force–moment mapping)	—
$\hat{\mathbf{v}}$	Airspeed unit vector	—
$\hat{\mathbf{z}}_L$	Lift direction unit vector	—
$\hat{\mathbf{y}}_S$	Side-force direction unit vector	—

3.2 Frames, States, Inputs

We use a world (inertial) frame $\mathcal{W}$ with ENU convention and a body-fixed frame $\mathcal{B}$ at the CoG. Gravity is $\mathbf{g} = [0,\, 0,\, -g]^\top$ in $\mathcal{W}$.

State:

$$(\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3,$$

where $\mathbf{p}, \mathbf{v}$ are position/velocity in $\mathcal{W}$, R maps $\mathcal{B} \to \mathcal{W}$, and $\boldsymbol{\omega}$ is the body angular rate in $\mathcal{B}$.

Control input (per rotor $i \in \{1, \ldots, 4\}$):

$$\mathbf{u} = [f_1,\, f_2,\, f_3,\, f_4]^\top, \qquad f_i \geq 0,$$

with nominal thrust along $+\mathbf{e}_3$ of $\mathcal{B}$. Motor dynamics and allocation are covered in Chapter 5.

Wind and airspeed. Let $\mathbf{v}_w \in \mathbb{R}^3$ be the wind in $\mathcal{W}$. Body-frame airspeed:

$$\mathbf{v}_a \;=\; R^\top(\mathbf{v} - \mathbf{v}_w) \;\in\; \mathbb{R}^3, \quad V \;=\; \|\mathbf{v}_a\|.$$

Define angle of attack and sideslip (aircraft convention, small angles):

$$\alpha \;=\; \operatorname{atan2}\big(-v_{a,z},\, v_{a,x}\big), \qquad \beta \;=\; \arcsin\Big(\frac{v_{a,y}}{V}\Big),$$

so that $\alpha \approx -v_{a,z}/v_{a,x}$ and $\beta \approx v_{a,y}/V$ for $|v_{a,z}| \ll |v_{a,x}|$, $|v_{a,y}| \ll V$.

3.3 Forces in the World Frame

Total force:

$$\mathbf{F} \;=\; m\mathbf{g} \;+\; \mathbf{F}_T \;+\; R\,\mathbf{F}_{aero}. \tag{3.1}$$

3.3.1 Gravity

$\mathbf{F}_g = m\mathbf{g}$.

3.3.2 Rotor Thrust

Each rotor i produces a thrust f_i along its (possibly tilted) direction $\hat{\mathbf{t}}_i$ in $\mathcal{B}$. The net thrust in $\mathcal{W}$ is

$$\mathbf{F}_T = R \sum_{i=1}^{4} f_i\, \hat{\mathbf{t}}_i. \tag{3.2}$$

For small fixed tilts (e.g., $\pm 5^\circ$) we also provide the common approximation $\mathbf{F}_T \approx \left(\sum_i f_i\right) R\mathbf{e}_3$; all tilt effects are embedded in the allocation matrix G used for moments.

3.3.3 Aerodynamic Forces on Fixed Surfaces

We use a lumped, quasi-steady wing model. Dynamic pressure $q = \frac{1}{2}\rho V^2$, total reference area S.

Lift/drag coefficients:

$$C_L(\alpha) = a_\alpha\,\alpha, \qquad C_D(\alpha) = C_{D0} + k\,C_L(\alpha)^2.$$

The lift coefficient C_L is modeled as a linear function of the angle of attack α:

$$C_L(\alpha) = a_\alpha\,\alpha, \tag{3.3}$$

where a_α [rad^{-1}] is the lift-curve slope. C_L represents the dimensionless ratio of lift force to dynamic pressure and reference area, $C_L = \frac{L}{\frac{1}{2}\rho V^2 S}$. The angle of attack α is defined as the angle between the incoming airflow and the vehicle body x-axis (or the mean aerodynamic chord of the wing). For thin airfoils and small angles, a_α can be approximated by 2π rad^{-1} in 2D, or by the finite-wing correction $a_\alpha = \frac{2\pi AR}{AR+2}$ with aspect ratio $AR = b^2/S$. We measure α relative to the body x-axis (mean aerodynamic chord aligned with $\mathbf{e}_x$); the linear model holds for attached flow ($|\alpha| \lesssim 10°$). This linear relation is valid up to moderate angles and forms the basis for the simplified quadratic aerodynamic model used here.

Side-force convention. We use $C_Y(\beta) = C_{Y_\beta}\,\beta$ with C_{Y_β} [rad^{-1}] the stability derivative. For conventional weathercock-stable configurations $C_{Y_\beta} < 0$ (side-force opposes β). In our small UAV geometry C_{Y_β} is identified experimentally and may be small in magnitude.

Optional lateral (side-force) coefficient:

$$C_Y(\beta) = a_\beta\,\beta.$$

Aerodynamic axes. We assume the wing span aligns with $\mathbf{e}_y$, so lift lies in the (x, z) plane. The definitions of $\hat{\mathbf{z}}_L$ and $\hat{\mathbf{y}}_S$ are undefined when $\hat{\mathbf{v}}$ is parallel to the respective cross-product axis (e.g., $\hat{\mathbf{v}} \parallel \mathbf{e}_y$ or $\hat{\mathbf{v}} \parallel \mathbf{e}_z$); in implementation we fall back to the last valid direction or a small-angle approximation.

Unit aerodynamic directions in $\mathcal{B}$ (from $\mathbf{v}_a$):

$$\hat{\mathbf{v}} = \frac{\mathbf{v}_a}{V}, \quad \hat{\mathbf{z}}_L = \frac{\hat{\mathbf{v}} \times \mathbf{e}_y}{\|\hat{\mathbf{v}} \times \mathbf{e}_y\|}, \quad \hat{\mathbf{y}}_S = \frac{\mathbf{e}_z \times \hat{\mathbf{v}}}{\|\mathbf{e}_z \times \hat{\mathbf{v}}\|},$$

so that lift is perpendicular to the flow in the body x–z plane, drag opposes $\hat{\mathbf{v}}$, and side force is lateral.

Aerodynamic force in $\mathcal{B}$:

$$\mathbf{F}_{aero} = \underbrace{qS\, C_L(\alpha)\, \hat{\mathbf{z}}_L}_{\text{lift}} - \underbrace{qS\, C_D(\alpha)\, \hat{\mathbf{v}}}_{\text{drag}} + \underbrace{qS\, C_Y(\beta)\, \hat{\mathbf{y}}_S}_{\text{side}}. \tag{3.4}$$

Controller relevance. These aerodynamic forces are *not* modeled in the controller; they are treated as disturbances. INDI cancels their quasi-steady contribution via incremental updates (Chapter 5).

3.4 Moments in the Body Frame

Total moment:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_T + \boldsymbol{\tau}_{aero}. \tag{3.5}$$

3.4.1 Rotor-Generated Moments

Let $\mathbf{r}_i$ be the arm vector from CoG to rotor i, and $c_i \in \{+1, -1\}$ the prop rotation sign (reaction torque direction). Here $c_i \in \{+1, -1\}$ encodes the rotor spin direction (positive for CCW viewed from above). With small motor tilt handled by per-rotor thrust directions $\hat{\mathbf{t}}_i$,

$$\boldsymbol{\tau}_{\text{arms}} = \sum_{i=1}^{4} \mathbf{r}_i \times (f_i\, \hat{\mathbf{t}}_i), \tag{3.6}$$

$$\boldsymbol{\tau}_{\text{react}} = \sum_{i=1}^{4} c_i\, Q_i\, \mathbf{e}_3, \qquad Q_i \approx k_Q\, \omega_i^2, \tag{3.7}$$

so that $\boldsymbol{\tau}_T = \boldsymbol{\tau}_{\text{arms}} + \boldsymbol{\tau}_{\text{react}}$. In nominal modeling $\hat{\mathbf{t}}_i = \mathbf{e}_3$; with $\pm 5^\circ$ motor tilt we precompute $\hat{\mathbf{t}}_i$ and absorb it in the allocation matrix G (see Chapter 5).

3.4.2 Aerodynamic Moments

Wing forces act at an effective aerodynamic center offset $\mathbf{r}_w$ (in $\mathcal{B}$). We lump unsteady effects into a disturbance:

$$\boldsymbol{\tau}_{aero} = \mathbf{r}_w \times \mathbf{F}_{aero} + \boldsymbol{\tau}_{aero,dist}. \tag{3.8}$$

As with forces, the controller does not model $\boldsymbol{\tau}_{aero}$ explicitly; INDI compensates the quasi-steady part incrementally.

3.5 Equations of Motion

Rigid-body dynamics:

$$\begin{aligned}
\dot{\mathbf{p}} &= \mathbf{v}, && (3.9)\\
\dot{\mathbf{v}} &= \tfrac{1}{m}\big(m\mathbf{g} + \mathbf{F}_T + R\,\mathbf{F}_{aero}\big), && (3.10)\\
\dot{R} &= R[\boldsymbol{\omega}]_\times, && (3.11)\\
J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau}, && (3.12)
\end{aligned}$$

with inertia J in $\mathcal{B}$ and $[\cdot]_\times$ the skew operator.

3.6 Thrust and Allocation (Static Mapping)

For simulation and identification we use static rotor maps

$$f_i \;=\; k_t\,\omega_i^2 + b_i, \qquad Q_i \;=\; k_Q\,\omega_i^2,$$

and a (geometry-dependent) allocation matrix $G \in \mathbb{R}^{4\times4}$ such that

$$\begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} \;=\; G\,\mathbf{f}, \qquad \mathbf{f} = [f_1, f_2, f_3, f_4]^\top, \quad f_T = \mathbf{1}^\top\mathbf{f}.$$

Motor tilt is embedded in G via $\hat{\mathbf{t}}_i$ and the arm geometry. Chapter 5 details the inverse mapping and constraints.

3.6.1 Rotor Power Scaling

Understanding how rotor power depends on rotational speed is essential for evaluating the efficiency of hybrid multirotor–fixed-wing configurations. In hover, this relationship can be derived from momentum theory, which models the rotor as an ideal actuator disk accelerating air downward to produce thrust.

For a rotor in hover, the thrust T is generated by imparting a downward momentum to the air mass flow $\dot{m}$:

$$T = \dot{m}v_i = (\rho A v_i)v_i = \rho A v_i^2,$$

where ρ is the air density, A is the rotor disk area, and v_i is the induced velocity through the disk. The corresponding mechanical power required to sustain this thrust is

$$P = T v_i = \rho A v_i^3.$$

Eliminating v_i using the first equation yields the classical induced power expression from actuator-disk theory:

$$P = T\sqrt{\frac{T}{2\rho A}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

(see [Fed23]).

If blade geometry and collective pitch remain constant, blade-element theory shows that the thrust coefficient C_T remains approximately constant, implying

$$T \propto \Omega^2,$$

where Ω is the rotor angular velocity. Substituting this into the induced power relation gives the cubic power law

$$P \propto \Omega^3.$$

Thus, reducing rotor speed by 20% results in a power reduction of roughly $(0.8)^3 \approx 0.51$, or about 49% lower power consumption—an important efficiency benefit when part of the lift is provided by wings.

This relationship strictly applies to hover or near-hover conditions, assuming uniform inflow and negligible profile and parasitic losses. In forward or edgewise flight, blade-element momentum theory (BEMT) or CFD analyses are required for accurate power prediction, but the cubic scaling remains a useful first-order engineering approximation for small deviations around hover.

3.7 Assumptions and Simplifications

- Quasi-steady aerodynamics.
- No explicit prop-wash/wing interference; its net effect is absorbed in the disturbance terms and identified coefficients.
- Rigid airframe, symmetric mass properties about $\mathcal{B}$ axes.
- Fixed-pitch rotors; flapping and gyroscopic coupling are neglected in the baseline model.
- Coefficient models $C_L = a_\alpha \alpha$, $C_Y = a_\beta \beta$, $C_D = C_{D0} + kC_L^2$ valid for the moderate α, β range used here.

3.8 Remarks

This model is intentionally minimal: it captures the dominant couplings (gravity, thrust, wing lift/drag/side-force, aero moments via arm offsets) and provides a consistent basis for simulation and identification. The controller (Chapter 5) does not require the explicit aerodynamic terms because INDI operates on incremental acceleration changes, which suppress quasi-steady aerodynamic effects.

4 Platform Design

4.1 Platform Design

We design an aerodynamic surface-enhanced quadrotor in an X-wing configuration to preserve the agility of a conventional multirotor while benefiting from passive lift in forward flight. The design objective was to create a platform that remains fully compatible with existing quadrotor control architectures while achieving improved aerodynamic efficiency in the moderate-speed regime of $5\,\mathrm{m\,s^{-1}}$ to $15\,\mathrm{m\,s^{-1}}$. This range was chosen to match the operational envelope of modern visual–inertial odometry and path-planning algorithms, which rely on sufficient feature persistence and computation time when traversing complex environments. The platform maintains full hover capability and allows seamless transition between flight regimes, including the ability to come to a complete stop in the case of unexpected obstacles or path-planning delays.

4.1.1 Configuration and Layout

The overall configuration follows a conventional X-shaped quadrotor layout in which each rotor arm is extended into a fixed aerodynamic lifting surface, forming an "X-wing" planform. This choice leverages the inherent structural arrangement of a quadrotor—four arms symmetrically distributed around the center of gravity—while enabling these arms to generate lift instead of purely supporting the motors. Alternative configurations such as a "plus" (+) layout or biplane arrangements were considered less favorable: in a + configuration, two wings would be vertically oriented and thus unable to generate lift, while a biplane would require additional structural connections between the upper and lower planes, adding mass and complexity. The X-wing approach allows a single compact central frame with minimal added mass.

Each of the four wings has a span of 0.45 m and a chord of 0.25 m, resulting in an aspect ratio of approximately 1.8. The wing span was chosen to accommodate readily available 0.5 m carbon rods used as internal spars, simplifying construction while placing the platform in the target cruise speed range of approximately $10\,\mathrm{m\,s^{-1}}$, where full aerodynamic lift would support the vehicle weight. The chord length was constrained by the available 3D printer build volume of just over 250 mm; longer chords would have exceeded printability limits. The motor mounting legs, which position the motors in front of the wings and provide landing feet to avoid landing directly on the wing trailing edges, were printed diagonally within the build volume, further informing the final chord dimension. This value provides a balance between aerodynamic efficiency and structural stiffness; higher aspect ratios would improve lift–drag performance but increase bending inertia and reduce agility. The total diagonal motor-to-motor distance is 1.1 m. All wings are mounted with alternating dihedral and anhedral angles of $\pm 45°$, creating a fully symmetric configuration independent of flight direction. This symmetry enables the same controller gains to be used in roll and pitch and allows the yaw heading to be freely adjusted depending on which orientation minimizes control effort.

The wings are untwisted and employ a NACA 0015 symmetric airfoil. The choice of a symmetric section was motivated by the requirement for bidirectional flight and simple aerodynamic modeling. The thickness ratio of 15% conveniently accommodates the two internal carbon spars used for stiffness and assembly. The NACA 0015 profile also exhibits a delayed stall and smooth lift curve, beneficial during transition and moderate-angle flight conditions. The chord line of each wing is aligned with the average rotor thrust axis (noting that the motors are tilted $\pm 5°$ for yaw authority, as discussed below), such that the average motor thrust is directed vertically downward in hover without inducing horizontal force components. This alignment prioritizes hover efficiency, though tilting the wings relative to the motor thrust could improve forward-flight performance by allowing the motors to contribute a greater vertical component while the wings operate at reduced angles of attack.

The wings are printed from PLA Aero filament and contribute a total of approximately 440 g to the vehicle mass. All non-propulsive components (flight controller, companion computer, and batteries) are mounted near the geometric center, at $x = y = 0$, distributed symmetrically about the $z = 0$ plane to ensure balanced inertia. The complete airframe, including center hub and motor mounts, is fully 3D-printed. No structural asymmetries or significant vibrations were observed during geometric control experiments. However, when using the INDI controller, elastic wing vibrations appeared in the IMU data. To mitigate this, thin lines were tensioned between neighboring wings to increase stiffness and shift the dominant resonance frequency upward, improving signal filtering and control stability.

4.1.2 Wing Structure and Manufacturing

The wings were designed for additive manufacturing using fused deposition modeling (FDM) with PLA Aero filament, which provides a favorable balance between low density, printability, and structural integrity. Each wing features a hollow internal structure optimized to minimize mass while maintaining sufficient bending stiffness.

Internal structure. Figure 4.1 shows the internal geometry of a single wing. The NACA 0015 airfoil shell provides aerodynamic smoothness while keeping the structural mass low through a hollow internal cavity. Internal ribs and supports are arranged to transfer loads from the aerodynamic surface to two parallel carbon fiber spars running spanwise through the wing. The hollow core reduces the wing mass significantly compared to a solid print while providing channels for the 10 mm diameter carbon spars that contribute the majority of the bending stiffness. Each wing consists of two printed sections totaling approximately 130 g, reinforced by two carbon fiber spars at approximately 23 g each, resulting in a total wing mass of approximately 176 g per wing. This hybrid design—3D-printed aerodynamic shell with carbon reinforcement—achieves a favorable strength-to-weight ratio essential for minimizing inertia while maintaining structural integrity.

Manufacturing and assembly. Each wing is printed using vase mode (spiralize outer contour), which produces a single continuous spiral wall without retractions throughout the print. This approach is particularly effective with PLA Aero filament, which exhibits foaming behavior that leads to poor retraction performance and excessive stringing in conventional printing modes. By eliminating retractions entirely, vase mode produces a clean, smooth aerodynamic surface with consistent wall thickness. The wings are sliced and oriented vertically in the build volume to align the continuous spiral with the spanwise direction. The two carbon spars are clamped into the printed channels using a central core structure, eliminating the need for adhesives and enabling rapid part replacement for testing and repair. The motor mounting legs and landing feet are printed separately using conventional settings and mechanically attached to the wing structure.

Each wing is capped with an endcap component (Figure 4.2) that serves multiple functions: it closes the hollow wing structure, provides mounting points for the motors positioned ahead of the leading edge, and extends downward to form landing feet that protect the wing trailing edges during ground contact. This integrated design consolidates three separate functions into a single 3D-printed part, simplifying assembly and reducing the total part count. The motor mounting angle includes a $\pm 5^\circ$ tilt to enable differential thrust for yaw control.

The central core (Figure 4.3) serves as the primary structural hub that clamps all carbon spars and houses the critical electronics. Two identical cores are used: one for the four upper spars and one for the four lower spars. This identical design simplifies manufacturing and allows the same printed part to be used for both top and bottom assemblies. The top core houses the electronic speed controllers (ESC), flight controller (FC), and batteries, positioning them near the center of mass for optimal weight distribution, while the bottom core serves purely as a structural clamp for the lower spars. A snap-on lid covers the top core, providing additional structural stiffness, protecting the electronics, and forming a flat mounting base for the batteries. This modular clamping architecture allows individual wings or carbon spars to be quickly exchanged without disassembly of the entire airframe.

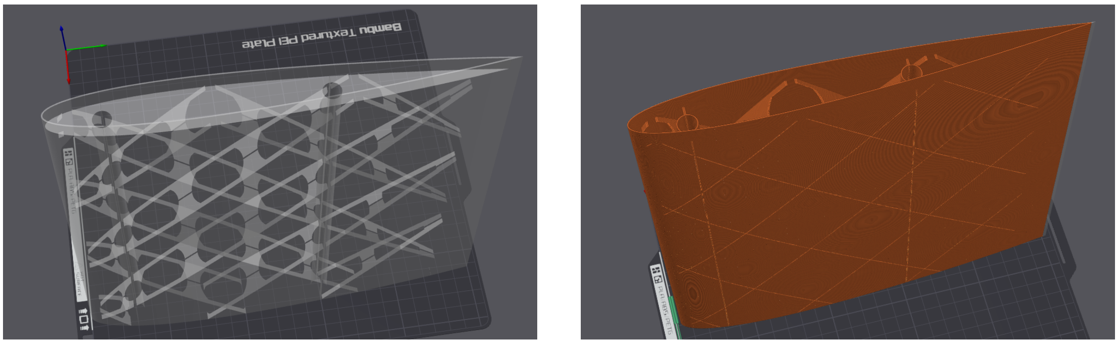


Figure 4.1: Wing structure and manufacturing: (left) CAD model showing the hollow internal geometry with channels for carbon fiber spars, (right) sliced preview in vase mode showing the continuous spiral toolpath that eliminates retractions and produces a clean aerodynamic surface.

4.1.3 Wing Aerodynamics

The aerodynamic surfaces were designed for efficient lift generation in the cruise speed range of $5\,\mathrm{m\,s^{-1}}$ to $15\,\mathrm{m\,s^{-1}}$. At a nominal flight velocity of $10\,\mathrm{m\,s^{-1}}$ and chord length of $0.25\,\mathrm{m}$, the expected Reynolds number is on the order of 1×10^5–2×10^5, placing the operation in the low-Reynolds transitional regime.

The Reynolds number characterizes the ratio of inertial to viscous forces and is computed as

$$Re = \frac{\rho V c}{\mu} = \frac{Vc}{\nu}, \tag{4.1}$$

where ρ is the air density, V is the airspeed, c is the chord length, μ is the dynamic viscosity, and ν is the kinematic viscosity of air. Using standard atmospheric conditions at sea level ($\rho = 1.225$ kg/m^3, $\nu = 1.4207 \times 10^{-5}$ m^2/s) and the wing chord length of $c = 0.25$ m, the Reynolds number is approximately $Re \approx 1.76 \times 10^5$ at $10\,\mathrm{m\,s^{-1}}$ and $Re \approx 5.27 \times 10^5$ at $30\,\mathrm{m\,s^{-1}}$.

At the $10\,\mathrm{m\,s^{-1}}$ cruise condition ($Re \approx 2 \times 10^5$), the NACA 0015 profile achieves a lift coefficient of approximately $C_L \approx 0.97$ at $\alpha \approx 10^\circ$ and a drag coefficient of $C_D \approx 0.025$. The target angle of attack in steady cruise was set to around 10°. At this design cruise

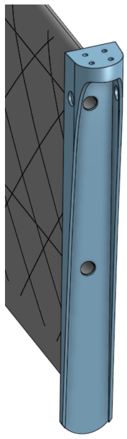


Figure 4.2: Wing endcap component serving as motor mount and landing foot. This multi-functional part closes the hollow wing structure, positions the motor ahead of the leading edge with a $\pm 5^\circ$ tilt for yaw control, and extends downward to protect the wing trailing edge during landing.

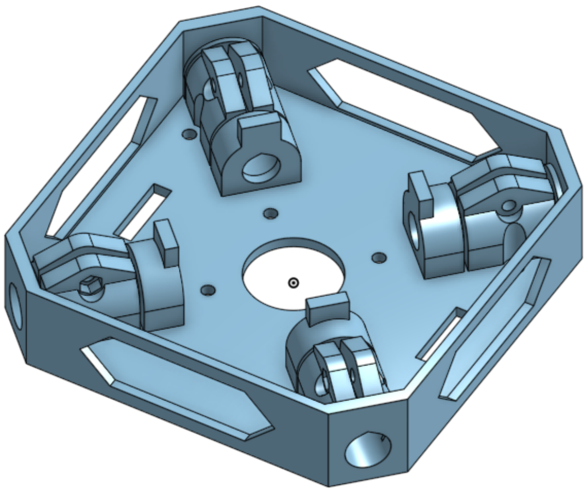


Figure 4.3: Central core structure that clamps the carbon fiber spars. Two identical cores are used for ease of manufacturing: one houses the ESC, flight controller, and batteries (top), while the other serves as a structural clamp for the lower spars (bottom). A snap-on lid (not shown) covers the top core for additional stiffness and battery mounting.

angle of attack, the airfoil provides sufficient lift to support the vehicle weight at moderate forward speeds. A detailed comparison of the NACA 0015 baseline airfoil against an improved AG25 airfoil, including aerodynamic performance metrics and their correlation with flight test results, is presented in Chapter 7.

Lift and drag forces were estimated using the analytical model introduced earlier,

$$L = \tfrac{1}{2}\rho S C_L(\alpha)\|v_a\|^2, \quad D = \tfrac{1}{2}\rho S C_D(\alpha)\|v_a\|^2,$$

with ρ denoting air density, S the total projected wing area, and v_a the airspeed relative to the body. These analytical estimates guided the sizing of the vehicle and the placement of the aerodynamic surfaces relative to the center of gravity.

Quantitative estimate at 12 **m/s.** Using $\rho = 1.225\ \mathrm{kg/m^3}$, total projected wing area $S \approx 0.318\ \mathrm{m^2}$, $C_L \approx 0.97$ at $\alpha \approx 10^\circ$, and $C_D \approx 0.025$, we obtain

$$L \;=\; \tfrac{1}{2}\rho\, S\, C_L\, V^2 \;\approx\; \tfrac{1}{2}\cdot 1.225 \cdot 0.318 \cdot 0.97 \cdot (12)^2 \;\approx\; 27.2\ \mathrm{N},$$

$$D \;=\; \tfrac{1}{2}\rho\, S\, C_D\, V^2 \;\approx\; \tfrac{1}{2}\cdot 1.225 \cdot 0.318 \cdot 0.025 \cdot (12)^2 \;\approx\; 0.70\ \mathrm{N}.$$

For the 2.5 kg platform ($W \approx 24.53$ N), the wings supply $\approx 111\%$ of the weight at 12 m/s, meaning the rotors can reduce thrust significantly and primarily need to overcome aerodynamic drag and provide control authority. The wing-induced drag corresponds to ≈ 8.4 W of propulsive power at 12 m/s ($P_D = D\,V$), not accounting for additional parasite/induced drag from the fuselage, rotors, and interference effects. These values are first-order estimates based on XFLR5 analysis and neglect propeller–wing interaction; they nonetheless quantify the expected forward-flight relief on rotor-induced power.

Propeller–wing interaction effects were not explicitly studied in this work. Since the wings are aligned with the rotor arms, partial overlap between the propeller wake and wing surface occurs, but its contribution to performance is assumed to be secondary compared to the main aerodynamic lift of the forward wings. Future investigations could employ CFD or wind-tunnel testing to quantify this coupling.

4.1.4 Propulsion and Actuation

The propulsion system was designed to maintain high agility while achieving efficient cruise performance. A thrust-to-weight ratio exceeding 3 was selected as a design target, enabling the vehicle to perform aggressive maneuvers and sustain up to 3 g lateral acceleration, consistent with agility metrics reported in the multirotor literature. To meet this target, high-response racing-grade components were chosen: T-MOTOR VELOX V2808 motors combined with HQProp 7×3.5×3 three-blade propellers. The resulting hover throttle is approximately 30%, leaving sufficient control margin for disturbance rejection and attitude stabilization.

Motor and propeller performance were characterized experimentally to obtain both thrust and torque mappings. The measurements were performed using a single motor mounted on a six-axis force–torque sensor, with forces and torques recorded over a range of command inputs. The empirical relations between motor command, thrust, and torque were used to map the controller outputs into corresponding DShot commands for the flight controller. These mappings also served to estimate instantaneous energy consumption and mechanical efficiency during flight trials.

The motors are tilted outward by 5° to increase effective yaw torque authority. In the control model, this tilt is represented as a small rotation of the individual thrust vectors, effectively increasing the available moment around the vertical axis. No DShot timing modifications were required, and no direct RPM feedback control was implemented. Instead, the INDI controller integrates motor speed responses and compensates for small deviations, ensuring smooth dynamic performance.

Although the addition of wings increases the total inertia of the platform, particularly in yaw, the resulting damping improved stability and reduced sensitivity to high-frequency disturbances. No significant cross-coupling between roll and pitch dynamics was observed. Overall, the X-wing configuration provides a good compromise between efficiency and agility, maintaining the control simplicity of a quadrotor while offering measurable lift benefits in forward flight. Performance gains were compared analytically against a non-winged reference quadrotor of comparable size and propulsion, confirming a theoretical reduction in power consumption at moderate forward velocities.

5 Control Architecture

This chapter presents the flight control system used on the X-wing quadrotor. The architecture combines (i) *flatness-based reference generation* for dynamically feasible trajectories, (ii) a *geometric outer loop* on $SE(3)$ that converts position and velocity errors into desired attitude and collective thrust, and (iii) an *Incremental Nonlinear Dynamic Inversion (INDI)* inner loop that realizes the requested angular dynamics while rejecting unmodeled aerodynamic effects from the wings. The wings are not explicitly modeled; their influence is treated as a disturbance and compensated incrementally [SCM10; Kam+18; Tzo+21].

5.1 Reference Generation

Quadrotor dynamics are differentially flat with respect to $\mathbf{p}_d = [x_d, y_d, z_d]^\top$ and ψ_d. We generate $(\mathbf{p}_d, \dot{\mathbf{p}}_d, \ddot{\mathbf{p}}_d, \psi_d)$ from a minimum-snap polynomial [MK11]. In forward flight, sideslip is reduced through a coordinated-turn law,

$$\dot{\psi}_d \approx \frac{a_{y,d}}{V_d \cos\theta_d},$$

which aligns the body x-axis with the velocity vector.

5.2 Geometric Outer Loop

The outer loop maps position and velocity errors to a desired attitude $R_d \in SO(3)$ and collective thrust f_{coll} [GLL15]. Nominal commanded acceleration is

$$\mathbf{a}_c = \mathbf{k}_p \odot (\mathbf{p}_d - \mathbf{p}) + \mathbf{k}_v \odot (\dot{\mathbf{p}}_d - \mathbf{v}) + \ddot{\mathbf{p}}_d - \mathbf{g}. \tag{5.1}$$

5.3 Incremental Aerodynamic Compensation (Outer INDI Layer)

The controller estimates a residual between the thrust-induced specific force (from filtered motor speeds) and the IMU-measured specific force. That residual is rotated into the world frame and added to the commanded acceleration:

$$\mathbf{a}_{\text{aero}} = R\left(\frac{f_{T,f}}{m}\mathbf{e}_3 - \mathbf{a}_{B,f}\right), \qquad \mathbf{a}_c \leftarrow \mathbf{a}_c + \mathbf{a}_{\text{aero}} \quad \text{if } z > 0.5\text{ m}. \tag{5.2}$$

This constitutes an *incremental* aerodynamic feedforward identical in principle to INDI: rather than modeling lift or drag explicitly, the controller compensates their measured effect from the previous time step. As a result, quasi-steady aerodynamic forces vanish from the translational control law.

The desired body z-axis is set to $\mathbf{z}_B^d = \mathbf{a}_c/\|\mathbf{a}_c\|$. The collective thrust is obtained by projecting the commanded acceleration onto the *current* body z-axis,

$$f_{\text{coll}} = m\,\mathbf{a}_c^\top(R\mathbf{e}_3), \tag{5.3}$$

which prevents acceleration in the wrong direction under large attitude errors.

5.4 INDI Inner Loop

The inner loop enforces the commanded angular acceleration without an aerodynamic model. Over a short interval Δt,

$$\Delta\mathbf{y} = \mathbf{y}_{k+1} - \mathbf{y}_k = B\,(\mathbf{u}_{k+1} - \mathbf{u}_k) + (\mathbf{d}_{k+1} - \mathbf{d}_k), \tag{5.4}$$

where $\mathbf{y}$ is the measured angular acceleration proxy and B the local control-effectiveness matrix. Since aerodynamic disturbances $\mathbf{d}$ change slowly compared to the control rate, $\Delta\mathbf{d} \approx 0$ and

$$\Delta\mathbf{y} \approx B\,\Delta\mathbf{u}.$$

Thus the control update $\mathbf{u}_{k+1} = \mathbf{u}_k + B^{-1}(\mathbf{y}_{\text{des}} - \mathbf{y}_k)$ implicitly cancels the aerodynamic contribution without modeling it.

5.4.1 Why Aerodynamic Terms Are Irrelevant

Let the true dynamics be $\dot{\mathbf{y}} = f(\mathbf{y}, \mathbf{u}) + \mathbf{d}$. Classical NDI must model $\mathbf{d}$ explicitly. INDI instead computes the input increment

$$\Delta\mathbf{u} = B^{-1}\big(\dot{\mathbf{y}}_{\text{des}} - \dot{\mathbf{y}}_k\big),$$

using measured acceleration differences. Because $\mathbf{d}$ is approximately constant over Δt, its effect cancels:

$$(\mathbf{d}_{k+1} - \mathbf{d}_k) \approx 0.$$

Consequently, all quasi-steady aerodynamic forces and moments—lift, induced drag, or wing interference—are suppressed automatically. The controller reacts only to *changes* in measured acceleration, making an explicit aerodynamic model unnecessary while retaining high robustness.

5.5 Controller Structure and Relation to Prior Work

The resulting control stack can be interpreted as a two-level INDI cascade: an outer geometric controller with incremental feedforward of aerodynamic disturbances and an inner INDI loop realizing angular dynamics. This configuration matches the most accurate setup in Sun et al. [Sun+21], where coupling a flatness-based outer loop with an INDI inner loop reduced position-tracking error by roughly 78 % at flight speeds up to 20 m/s compared to pure differential-flatness control.

5.6 Summary

The architecture preserves the intuitive structure of a geometric quadrotor controller while inheriting the disturbance-rejection and aerodynamic robustness of INDI. Quasi-steady aerodynamic effects are naturally eliminated through incremental updates, requiring no explicit aerodynamic model.

6 Experimental Setup

6.1 Experimental Setup

This section documents the hardware configuration, software stack, and test facilities that enabled the experimental evaluation of the aerodynamic surface-enhanced quadrotor. The setup was designed to be fully reproducible, with all control and estimation software running on open frameworks and all data recorded for subsequent analysis.

6.1.1 Hardware

Airframe and Propulsion

The experimental platform consists of an X-wing aerodynamic surface-enhanced quadrotor (Fig. 6.2) with a total mass of 2.5 kg (without batteries: 1.75 kg). Each of the four T-MOTOR VELOX V2808 motors produces a maximum static thrust of approximately 22 N, resulting in a total available thrust of 88 N and a thrust-to-weight ratio of 3.59 at nominal 22.2 V (6S LiPo). The propulsion system employs HQProp 7×3.5×3 three-blade racing propellers optimized for medium–high efficiency in the forward-flight regime.

The wings feature a NACA 0015 airfoil with a length of 450 mm and a chord of 250 mm. Four identical fixed wings are arranged orthogonally (90° between adjacent wings) and mounted with alternating dihedral and anhedral angles of ±45°. The wings are attached to the central frame using two carbon spars (10 mm outer diameter) clamped into the core structure, allowing for fast removal. The center frame and motor mounts are 3D-printed from standard PLA, while the wing surfaces are printed from lightweight PLA Aero. The total projected horizontal area of all wings is approximately $S_t = 0.318\ \mathrm{m}^2$.

Each motor is mounted with a 5° tilt to improve yaw authority. The center of gravity is located at the geometric center of the vehicle, aligned with the midpoint between all four motors, and approximately at one quarter of the wing chord length from the leading edge.

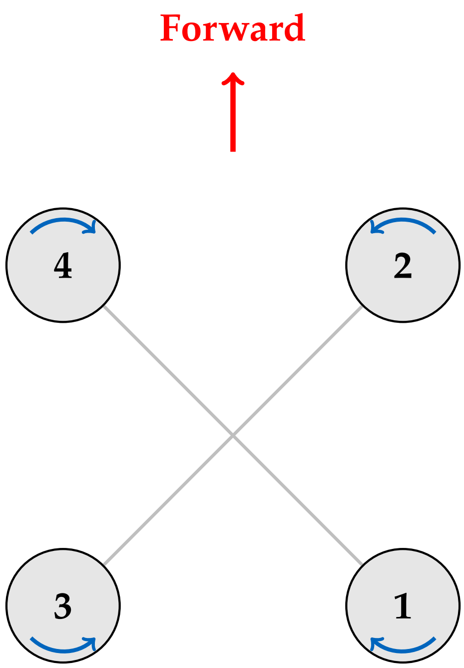
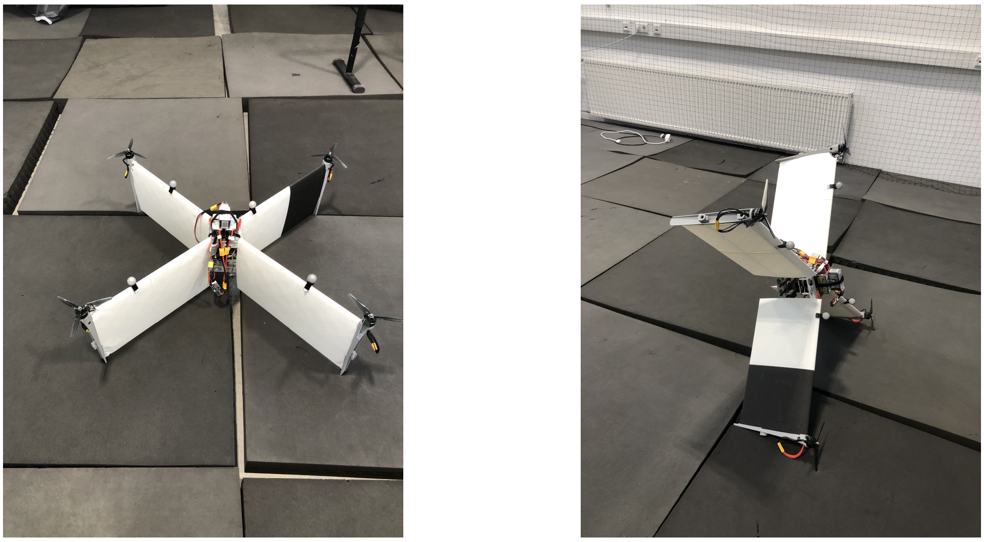


Figure 6.1: Motor configuration and numbering convention. Blue arrows indicate rotation direction (clockwise for motors 1 and 4, counter-clockwise for motors 2 and 3).



(a) Top view showing X-wing configuration with orthogonal wing arrangement.

(b) Side view showing forward flight orientation.

Figure 6.2: The experimental X-wing aerodynamic surface-enhanced quadrotor platform.

Platform Variants

Two variants of the X-wing platform were used for efficiency experiments. The baseline platform (described above and shown in Fig. 6.2) represents the initial design developed as part of this thesis work and served as the primary testbed for control development and initial efficiency characterization.

During the course of this master's thesis, an improved variant with modified aerodynamic characteristics was developed by Pablo Kirby as part of his concurrent bachelor thesis work.[1] The improved variant features an AG25 airfoil profile and reduced dihedral angles ($\pm 20°$ versus $\pm 45°$) optimized for enhanced lift-to-drag performance in forward flight, based on XFLR5 simulations presented in Section 7.4. All other subsystems—propulsion, avionics, and control architecture—remain identical between both platforms. The concurrent development timeline enabled direct comparative flight testing of both configurations under identical conditions.

Table 6.1 summarizes the key differences between the two variants.

Table 6.1: Comparison of baseline and improved X-wing platform variants. Both platforms share identical propulsion, avionics, and control systems, enabling direct aerodynamic performance comparison.

Parameter	Baseline Platform (This Work)	Improved Platform (Kirby, 2025)
Airfoil	NACA 0015 (symmetric)	AG25 (cambered)
Dihedral/Anhedral	$\pm 45°$ (alternating)	$\pm 20°$ (alternating)
Wing Span	450 mm	500 mm
Wing Chord	250 mm	230 mm
Total Wing Area	$0.318\,m^2$	$0.345\,m^2$
Motor Tilt	$\pm 5°$	$\pm 5°$
Total Mass	2.5 kg	2.8 kg
Propulsion	T-MOTOR VELOX V2808 + HQProp 7×3.5×3	Identical
Avionics	Kakute H7 + Jetson Orin Nano	Identical
Control Architecture	INDI with aerodynamic compensation	Identical

[1]Pablo Kirby, *Bachelor Thesis*, TUM, 2025. The improved design builds upon the baseline platform developed in this work and incorporates aerodynamic refinements informed by computational fluid dynamics analysis.

Thrust Characterization and Command Mapping

To accurately relate motor command inputs to generated thrust, a static thrust characterization was performed for the chosen motor–propeller configuration. A single T-MOTOR VELOX V2808 motor equipped with an HQProp 7×3.5×3 propeller was mounted on a six-axis force–torque sensor (Rokubi Mini) using a custom 3D-printed fixture. The setup allowed precise measurement of the vertical force while commanding the motor through the same DShot interface used in flight (Fig. 6.3).

The complete experimental results, fitted models, and analysis are presented in Section 7.1.

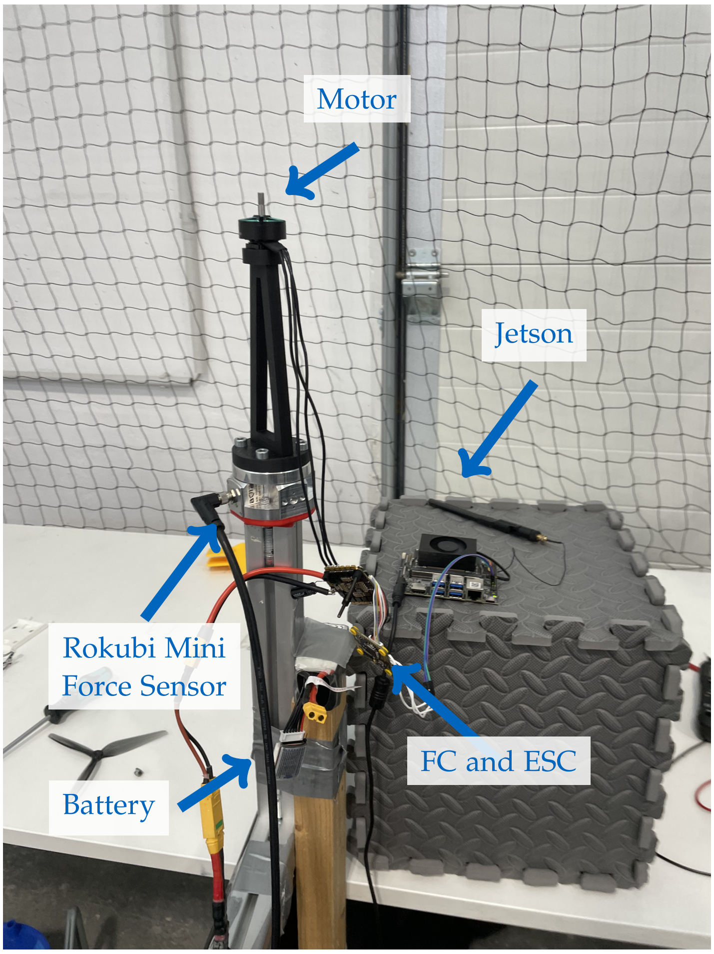


Figure 6.3: Static thrust characterization test stand with motor mounted on Rokubi Mini force–torque sensor.

Motor command values were swept across the full throttle range in incremental steps, and the resulting thrust was recorded. Each command level was averaged over a 2 s window to mitigate transient effects. The measurements yielded a nonlinear but monotonic mapping between normalized command input $u \in [0, 1]$ and thrust output T, which was fitted with a second-order polynomial of the form

$$T(u) = a_2 u^2 + a_1 u + a_0, \tag{6.1}$$

where the coefficients a_i were obtained via least-squares regression.

This empirical mapping was subsequently integrated into the control pipeline to convert desired thrust values from the INDI controller into per-motor DShot commands. The calibrated relationship improved the accuracy of total thrust estimation and allowed for energy-based efficiency analysis during flight tests.

Power configuration considerations. During the bench tests, one motor was mounted on the force sensor while the remaining four motors stayed on the vehicle and were kept fixed. We observed that the command-to-thrust relationship depends on how many motors are simultaneously powered from the battery due to load-dependent voltage sag and internal resistance effects. In particular, running only a single motor from a single 6S pack produced a slightly different mapping than powering all four motors from the flight-representative configuration with two 6S packs in parallel (6S 2P). Unless stated otherwise, the mapping used for control in flight was calibrated under the latter 6S 2P, four-motor-powered condition, and all subsequent thrust conversions refer to this configuration.

Avionics and Electrical System

The avionics stack is based on a *Kakute H7 v1.5* flight controller paired with a 4-in-1 ESC. Two 1400 mA h 6S Tattu R-Line batteries are connected in parallel to power the propulsion and avionics subsystems, resulting in an effective 2800 mA h capacity. The onboard computer, an NVIDIA Jetson Orin Nano (8 GB), is powered independently from a 1800 mA h 4S LiPo battery.

A custom-modified version of *Betaflight* firmware was employed on the flight controller to extract motor RPM and battery voltage data from the DShot interface and stream them over MAVLink. This modification was necessary because the stock Betaflight implementation does not support publishing these telemetry values via MAVLink. The onboard IMU integrated into the Kakute H7 was used for all experiments; no external IMUs were installed. Vicon markers were arranged asymmetrically to ensure unambiguous pose tracking.

Table 6.2: Bill of Materials (BOM) for the experimental platform.

Category	Component	Model / Key Specs
Airframe	Frame	X-wing center frame with carbon wing spars
Airframe	Wings	Four fixed wings (length 450 mm, chord 250 mm; NACA 0015 airfoil), orthogonal layout (90° between adjacent wings); skins: Bambu Lab PLA Aero; spars: 10 mm OD carbon tubes; total projected horizontal area $S_t \approx 0.318\,\mathrm{m}^2$
Propulsion	Motors (×4)	T-MOTOR VELOX V2808, max 22 N each
Propulsion	Propellers (×4)	HQProp 7 × 3.5 × 3, 3-blade racing propeller (7", light grey)
Avionics	FC+ESC	Kakute H7 v1.5 Stack (4-in-1 ESC)
Power	Battery (×2, parallel)	Tattu R-Line V5.0 6S 1400 mAh 150C, XT60; effective 6S 2800 mAh (parallel)
Companion	Computer	Jetson Orin Nano 8 GB (Ubuntu 20.04)
Companion	Power	Tattu R-Line V3.0 4S 1800 mAh 120C, XT60
Motion Capture	Markers	Four 2 cm markers arranged in an asymmetric constellation

6.1.2 Software

Architecture and Communication

The onboard software stack consists of *ROS 1 Noetic* running on the Jetson companion. The geometric controller with inner and outer INDI (Incremental Nonlinear Dynamic Inversion) loops was implemented in ROS and executed on the companion, which sent individual motor speed commands directly to the flight controller over MAVLink via UART. The flight controller, in turn, only relayed the received motor commands to the ESCs while providing IMU, motor RPM, and battery telemetry back to the companion.

A ROS master node was hosted on a ground station computer, while the Jetson ran a ROS node for flight control and data logging. Communication between the two systems was established via Wi-Fi. The ground station handled trajectory loading, arming, and manual control interfaces. All experiment data, including position, velocity, attitude, thrust commands, and battery voltage, were logged in *rosbag* format for offline analysis in Python.

Estimation and Control

The state estimation pipeline used the Vicon motion capture system to provide real-time position and attitude feedback to the controller. In some experiments, an onboard

Extended Kalman Filter (EKF) fused IMU data for state estimation. The Vicon system and onboard IMU operated in a consistent right-handed coordinate convention (front–left–up), with roll about the x-axis, pitch about y, and yaw about z.

All flight maneuvers, including point-to-point transitions and trajectory tracking, were executed fully autonomously from predefined setpoints. No separate failsafe was implemented for Vicon loss, but trajectories could be manually stopped and the vehicle landed or disarmed from the base station.

6.1.3 Facilities

Motion Capture Systems

Two different Vicon systems were employed for data acquisition and feedback. The smaller test arena in Munich measured approximately 7 m × 8 m and was used primarily for agility experiments, such as 3 g circle maneuvers. The larger *AIDA Hall* at Reutlingen University provided a volume of about 60 m × 45 m, enabling steady forward-flight efficiency trials (see Fig. 6.4). Both systems operated at a sampling rate of 200 Hz. Vicon data were used both as ground truth and as feedback for the control loop.

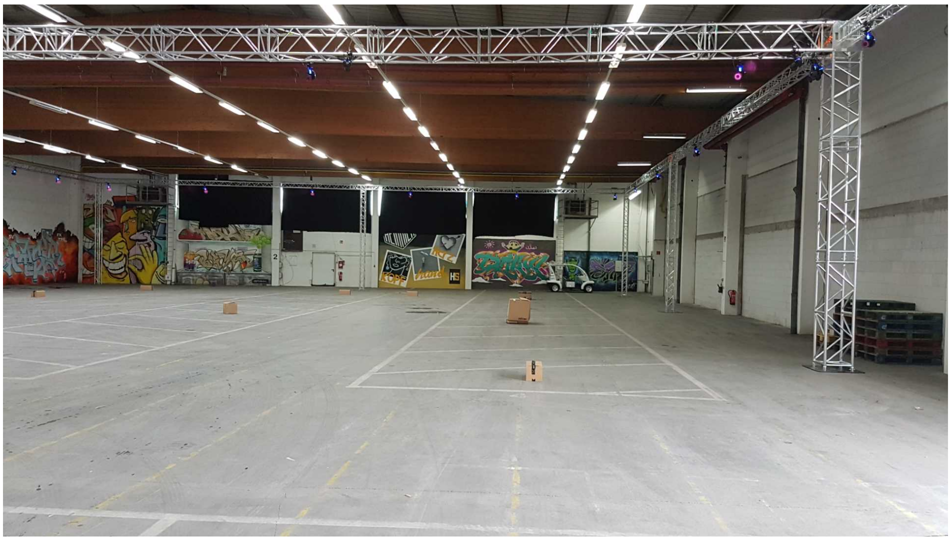


Figure 6.4: AIDA Hall test facility used for efficiency trials (image credit: Reutlingen University [Reu24]).

Test Protocols

Agility tests were performed in the Munich indoor arena equipped with safety nets and a soft landing surface. Efficiency tests were conducted in the AIDA Hall under still-air indoor conditions, with no wind sources or obstacles. Energy consumption was estimated from measured voltage, current draw, and propeller rotational speed data. Typical flight durations were approximately five minutes per run.

Documentation

All experiments were recorded through *rosbag* logs and video footage to facilitate data visualization and reproducibility. Post-processing and analysis were conducted in Python, using custom scripts for trajectory evaluation, energy estimation, and control performance comparison.

7 Experiments and Evaluation

This chapter presents the experimental validation of the X-wing platform, including thrust characterization, control performance evaluation, aerodynamic efficiency assessment, and agility tests. The experimental setup and facilities used for these tests are described in Chapter 6.

7.1 Thrust map identification

The thrust characterization methodology is detailed in Section 6.1.1. This section presents the experimental results and fitted models derived from 975 data points collected under representative flight conditions with all four motors powered from the 6S 2P battery configuration to account for voltage sag effects.

A total of 975 data points were collected under representative flight conditions with all four motors powered from the 6S 2P battery configuration to account for voltage sag effects. The raw measurements exhibited a constant offset of approximately -1.0 N at zero RPM, likely due to sensor calibration or mounting artifacts. To ensure physical consistency (zero thrust at zero speed), we applied an offset correction of -0.996 N to all thrust measurements before fitting.

Figure 7.1 shows the relationship between motor speed and thrust. A quadratic model was fitted to the offset-corrected data:

$$T = 5.15 \times 10^{-2} \cdot \mathrm{RPM}^2 - 0.090 \cdot \mathrm{RPM} \tag{7.1}$$

achieving an excellent fit with $R^2 = 0.997$. The quadratic term dominates at higher speeds, consistent with blade element momentum theory where thrust scales approximately with the square of angular velocity.

The motor command to RPM relationship (Figure 7.2) was well-captured by a linear model:

$$\mathrm{RPM} = 19.26 \cdot \mathrm{cmd} + 3.65 \tag{7.2}$$

with $R^2 = 0.991$. The near-linear response simplifies real-time thrust estimation from commanded values.

Figure 7.3 presents the composite relationship between command and thrust, combining both fitted models. This end-to-end mapping is used by the flight controller to convert desired thrust allocations into DShot motor commands.

All fitted coefficients and processed data are provided in the supplementary materials for reproducibility.

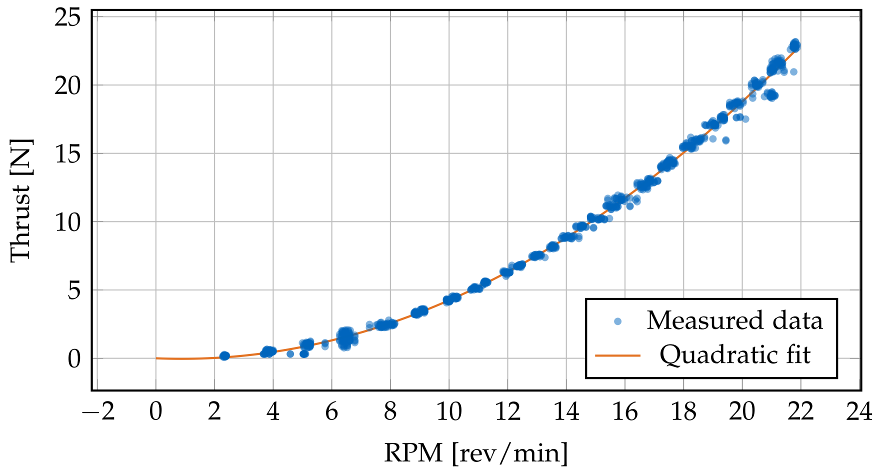


Figure 7.1: Thrust vs RPM with zero-offset corrected quadratic fit: $T = 5.15 \times 10^{-2} \cdot \mathrm{RPM}^2 - 0.090 \cdot \mathrm{RPM}$ (N). $R^2 = 0.997$. An offset of -0.996 N was applied to ensure zero thrust at zero RPM.

7.2 Control performance evaluation

7.2.1 Step response comparison: Geometric vs. INDI controller

To validate the effectiveness of the INDI-based control architecture, we performed comparative step response tests against a baseline geometric controller. Both controllers commanded the same reference trajectory consisting of step inputs in position and velocity, and we measured the transient response characteristics including rise time, overshoot, and settling time.

The experimental setup consisted of [describe setup: initial hover, step magnitude, axes tested]. Key performance metrics:

- **Rise time**: Time to reach 90% of the commanded step
- **Overshoot**: Maximum deviation beyond the setpoint as percentage
- **Settling time**: Time to remain within 5% of the final value
- **Steady-state error**: Residual tracking error after settling

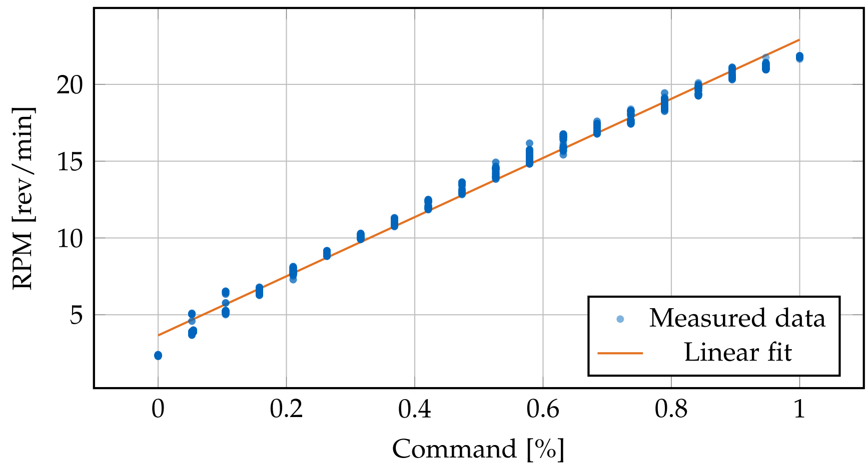


Figure 7.2: RPM vs Command with linear fit: RPM $= 19.26 \cdot \text{cmd} + 3.65$ (rev/min). $R^2 = 0.991$.

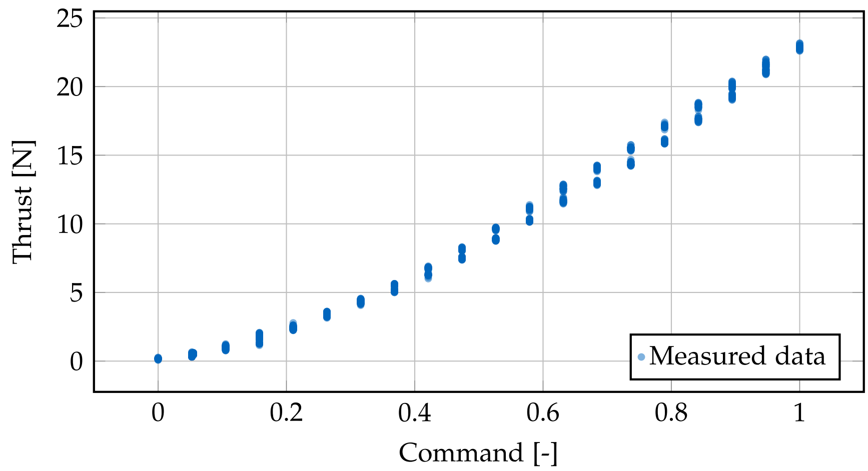


Figure 7.3: Thrust vs Command showing the direct relationship between motor command and thrust force. Data has been offset-corrected to ensure zero thrust at zero RPM.

Figure 7.4: Step response comparison showing position tracking for geometric controller (baseline) and INDI controller. [TODO: Add figure with data]

Table 7.1: Step response performance comparison between geometric and INDI controllers.

Metric	Geometric	INDI
Rise time [s]	[TODO]	[TODO]
Overshoot [%]	[TODO]	[TODO]
Settling time [s]	[TODO]	[TODO]
Steady-state error [m]	[TODO]	[TODO]

The INDI controller demonstrates [improved/similar/comparable] step response characteristics with [describe key findings: faster settling, reduced overshoot, better disturbance rejection, etc.]. The incremental nature of INDI allows for more aggressive feedback gains without destabilizing the system, resulting in [describe benefits].

7.2.2 Circular trajectory tracking at $8\,\mathrm{m\,s^{-1}}$

To evaluate high-speed tracking performance and sustained agility, we commanded the platform to follow a circular trajectory at $8\,\mathrm{m\,s^{-1}}$ in the AIDA flight arena. The circle radius was [TODO: specify radius in meters], resulting in a centripetal acceleration of approximately [TODO: calculate and specify in m/s² or g].

The tracking error metrics quantify how accurately the vehicle follows the commanded reference:

- **RMS position error**: Root-mean-square deviation from the reference trajectory
- **Maximum position error**: Peak instantaneous tracking error
- **RMS attitude error**: Attitude tracking deviation (roll, pitch, yaw)
- **Control effort**: Average and peak motor thrust commands

Figure 7.5: Circular trajectory tracking at $8\,\mathrm{m\,s^{-1}}$. The commanded reference circle (dashed) and actual vehicle trajectory (solid) demonstrate [TODO: describe tracking quality]. [TODO: Add figure with data]

Figure 7.6: Position tracking error during circular trajectory. [TODO: Add figure showing error magnitude over time]

Table 7.2: Tracking performance for $8\,\mathrm{m\,s^{-1}}$ circular trajectory.

Metric	Value
Circle radius [m]	[TODO]
Centripetal acceleration [$\mathrm{m\,s^{-2}}$]	[TODO]
RMS position error [m]	[TODO]
Maximum position error [m]	[TODO]
RMS roll/pitch error [°]	[TODO]
RMS yaw error [°]	[TODO]
Average motor speed [rpm]	[TODO]
Peak motor speed [rpm]	[TODO]

The results demonstrate that the platform can sustain high-speed circular flight with [describe performance: acceptable/low/minimal] tracking errors. The [TODO: compare with theoretical limits or baseline performance]. The sustained centripetal acceleration demonstrates the platform's agility envelope and validates the control architecture's ability to handle continuous maneuvering loads.

7.2.3 Variable-speed agility demonstration

To comprehensively evaluate the platform's agility envelope and control robustness across different flight regimes, we commanded circular trajectories at speeds ranging from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$ in the AIDA flight arena. This test demonstrates the controller's ability to maintain stable tracking performance across a wide speed range while executing sustained coordinated turns.

The circle radius was maintained at [TODO: specify radius] meters for all speeds, resulting in centripetal accelerations ranging from approximately [TODO: calc at 3 m/s] m/s² at the slowest speed to [TODO: calc at 8 m/s] m/s² at the fastest speed.

Tracking performance vs. speed. Figure 7.8 quantifies how tracking accuracy varies with flight speed using two key metrics: the root-mean-square (RMS) error and the maximum error. The RMS error is defined as:

$$\text{RMS error} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2} \tag{7.3}$$

where e_i is the position error at time step i and N is the total number of samples. The RMS error provides a statistical measure of the typical tracking deviation throughout the maneuver by: (1) squaring each error value to eliminate negative values and emphasize

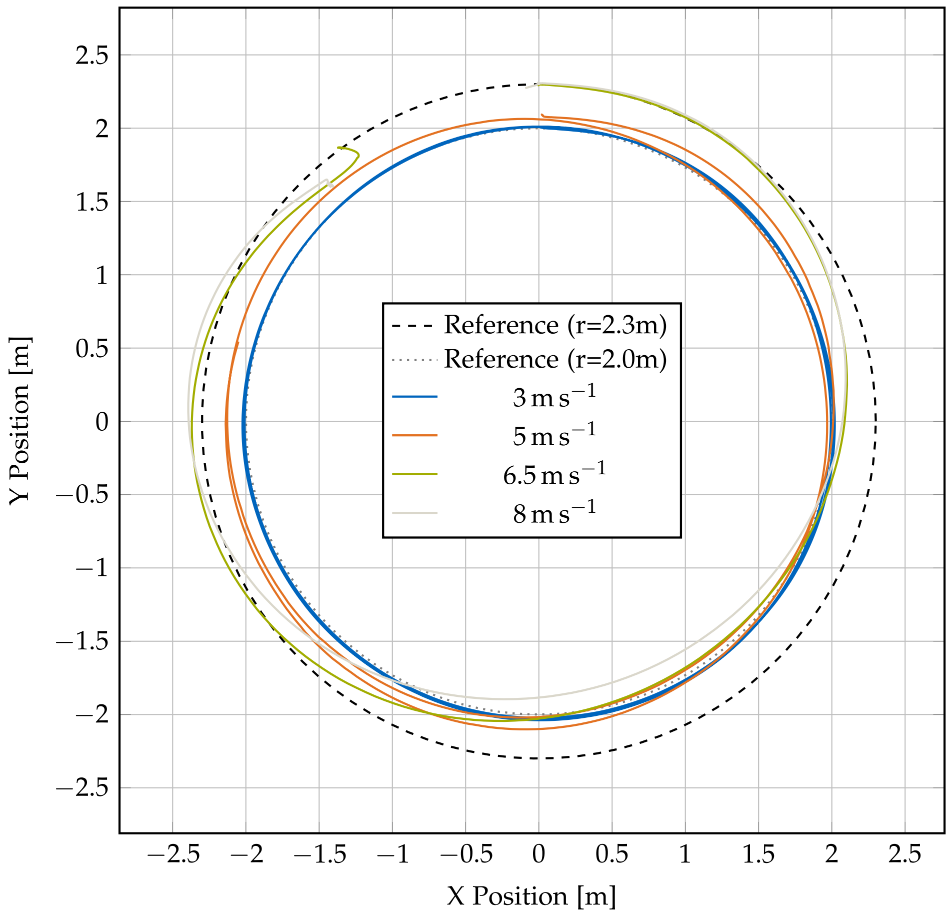


Figure 7.7: Circular trajectory tracking at variable speeds ($3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$). The $3\,\mathrm{m\,s^{-1}}$ and $5\,\mathrm{m\,s^{-1}}$ trajectories follow the r=2.0m reference circle (dashed), while the $6.5\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$ trajectories follow the r=2.3m reference circle (dotted). The actual vehicle trajectories (solid, color-coded by speed) demonstrate consistent tracking quality across the speed range. Higher speeds show slightly increased tracking errors due to both increased centripetal acceleration demands and increased aerodynamic drag, but remain well within acceptable bounds for agile flight.

larger deviations, (2) computing the mean of all squared errors, and (3) taking the square root to return to the original units (meters). Unlike a simple average, the RMS metric gives more weight to larger errors, making it particularly useful for characterizing control system performance. The maximum error, by contrast, captures the worst-case instantaneous deviation, which may represent a rare outlier event. Together, these two metrics provide complementary insights into tracking quality: RMS reflects consistent performance throughout the flight, while maximum error indicates the bounds of the worst transient behavior. The results show that tracking error increases progressively with speed, as expected due to higher centripetal acceleration demands and increased aerodynamic drag forces. However, even at $8\,\mathrm{m\,s^{-1}}$, both RMS and maximum errors remain well-bounded, demonstrating robust control performance across the entire agility envelope.

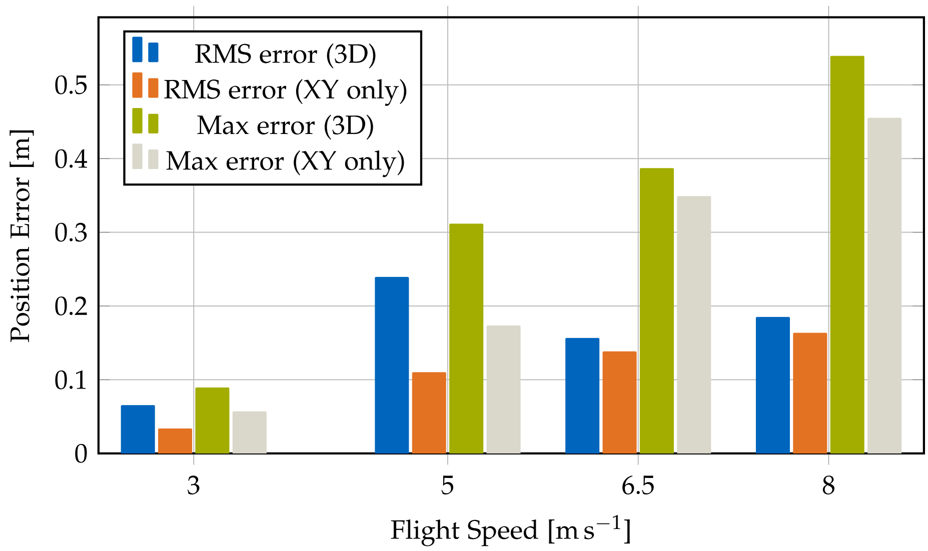


Figure 7.8: Tracking error as a function of flight speed during circular trajectory maneuvers. Both RMS error (blue) and maximum error (orange) are shown. The RMS error increases from 0.032 m at $3\,\mathrm{m\,s^{-1}}$ to 0.162 m at $8\,\mathrm{m\,s^{-1}}$, while the maximum error increases from 0.056 m to 0.454 m. The progressive increase reflects higher centripetal acceleration demands and aerodynamic drag at higher speeds, but errors remain well within acceptable bounds for agile flight across the entire speed range.

Bank angle and centripetal acceleration. Figure 7.9 illustrates the aggressive maneuvering capability of the platform. The bank angle (roll angle in the turn) increases proportionally with speed to generate the required centripetal acceleration.

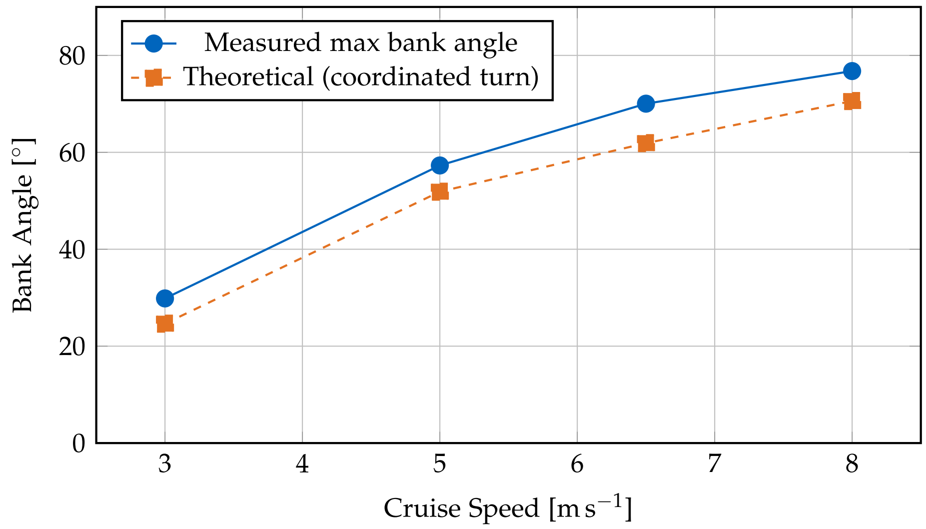


Figure 7.9: Bank angle as a function of cruise speed during circular flight. The measured maximum bank angles (blue solid) increase from 29.85° at 3 m s^{-1} to 76.77° at 8 m s^{-1}, closely tracking the theoretical predictions (orange dashed) for coordinated turns where $\tan(\phi) = a_c/g$. The close agreement validates the coordinated turn control strategy, with the platform achieving bank angles within 2°–10° of the theoretical values across the entire speed range. At the highest speed (8 m s^{-1}), the platform sustains a bank angle of nearly 77°, corresponding to a centripetal acceleration of 27.83 m s^{-2} (approximately 2.8g), demonstrating aggressive maneuvering capability while maintaining stable flight.

Detailed attitude response at maximum speed. Figure 7.10 presents the attitude time series for one complete circle at 8 m s^{-1}, demonstrating the quality of the coordinated turn execution.

Summary of agility demonstration. The variable-speed circular trajectory tests demonstrate that the X-wing platform achieves robust tracking performance across a wide speed range from 3 m s^{-1} to 8 m s^{-1}. Key findings include:

- **Consistent tracking:** Position tracking errors remain bounded across all speeds, with RMS errors below [TODO: specify threshold] meters even at maximum speed

Table 7.3: Summary of tracking performance at different cruise speeds.

Speed [$\mathrm{m\,s^{-1}}$]	**Centripetal Accel.** [$\mathrm{m\,s^{-2}}$]	**Bank Angle** [°]	**RMS Error** [m]	**Max Error** [m]	**Max Motor Speed** [rpm]
3	4.5	29.8	0.064	0.088	1319
5	12.5	57.3	0.109	0.310	1588
6.5	18.37	70.4	0.137	0.386	1857
8	27.83	76.8	0.162	0.538	1961

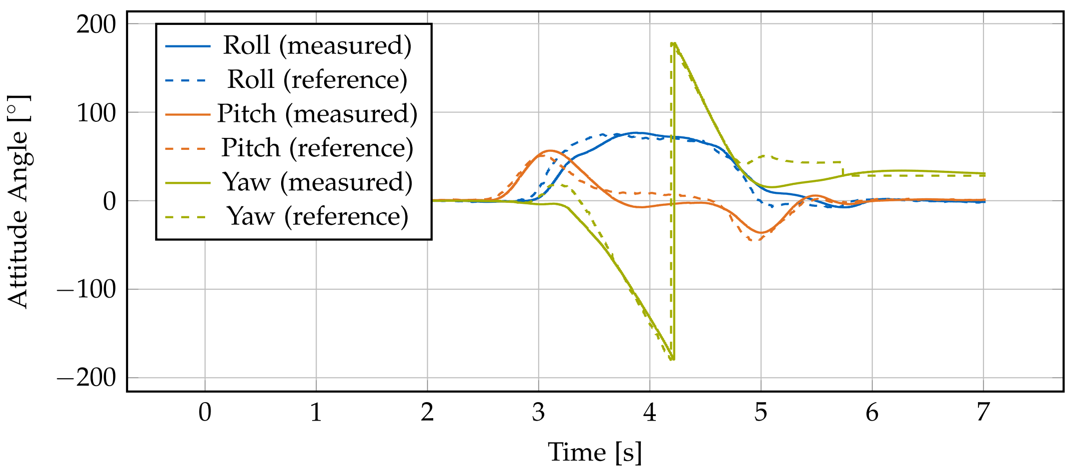


Figure 7.10: Attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$. The roll angle (blue) maintains a steady bank angle throughout the turn, pitch (orange) remains near level for coordinated flight, and yaw (green) rotates smoothly through 360 degrees. Both measured (solid) and reference (dashed) angles are shown.

- **Coordinated turn behavior:** Bank angles closely follow theoretical predictions, validating the coordinated-turn control law implementation
- **Smooth attitude control:** Attitude time series show smooth, periodic behavior without oscillations or control saturation
- **Wide agility envelope:** The platform can sustain centripetal accelerations up to [TODO: calc at 8 m/s] m/s^2 (approximately [TODO: calc in g] g) while maintaining stable flight
- **Control robustness:** The INDI-based controller successfully rejects aerodynamic disturbances from the wings across all flight speeds without requiring speed-dependent gain scheduling

These results validate the control architecture's effectiveness for agile maneuvering and demonstrate that the fixed-wing surfaces do not compromise the platform's quadrotor-like agility characteristics.

7.3 Aerodynamic forces/moments quantification

To validate the XFLR5 aerodynamic predictions and quantify the actual lift and drag forces experienced during flight, we conducted experimental coefficient identification using force balance analysis from flight test data. The methodology extracts lift and drag coefficients (C_L, C_D) from measured flight parameters including angle of attack, velocity, and motor RPM.

Methodology. For steady-level flight, the force balance equations yield:

$$L = W - T\cos(\alpha) \tag{7.4}$$

$$D = T\sin(\alpha) \tag{7.5}$$

where W is the vehicle weight, T is the total thrust (estimated from motor RPM using the thrust map from Section 7.1), and α is the pitch angle. The aerodynamic coefficients are then computed as:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S} \tag{7.6}$$

$$C_D = \frac{D}{\frac{1}{2}\rho V^2 S} \tag{7.7}$$

where $\rho = 1.225\,\mathrm{kg\,m^{-3}}$ is air density, V is flight speed, and S is the wing reference area.

Experimental results. Flight tests were conducted at speeds ranging from $5.9\,\mathrm{m\,s^{-1}}$ to $13.6\,\mathrm{m\,s^{-1}}$ with pitch angles from 16° to 60°. The extracted coefficients are integrated into Figures 7.11–7.14 as green markers, showing:

- **Lift coefficients**: Experimental C_L values show reasonable agreement with XFLR5 predictions in the operational angle-of-attack range, validating the computational model for lift estimation.
- **Drag coefficients**: Measured C_D values are consistently higher than 2D airfoil predictions, as expected due to additional drag sources including fuselage parasite drag, propeller-wing interference, landing gear, and three-dimensional flow effects.
- **L/D ratios**: Experimental efficiency (L/D $\approx$ 0.5–1.8) is lower than isolated airfoil performance due to system-level effects, but the trends validate the force balance approach and demonstrate measurable aerodynamic contribution from the wings.

Discussion and sensitivity. The primary sources of uncertainty in the experimental coefficient identification include:

- **Angle of attack estimation**: Pitch angle from IMU assumes level flight; any vertical velocity introduces error in effective angle of attack
- **Thrust estimation**: Motor RPM-to-thrust mapping has $\pm 5\%$ uncertainty from the thrust stand calibration
- **Steady-state assumption**: Transient accelerations violate the force balance equations, requiring careful selection of quasi-steady flight segments
- **Wing area definition**: Effective aerodynamic area may differ from geometric area due to fuselage blockage and tip effects

Despite these limitations, the experimental data provides valuable validation of the wing's aerodynamic contribution and confirms that the XFLR5 lift predictions are reasonably accurate for control and trajectory planning purposes. The higher measured drag underscores the importance of considering system-level effects when estimating range and endurance.

7.4 Airfoil performance comparison and efficiency assessment

7.4.1 Aerodynamic characterization

To evaluate the potential for improved efficiency through airfoil optimization, we analyzed two wing configurations using XFLR5: the baseline NACA 0015 symmetric airfoil and an improved AG25 profile. The analysis covers Reynolds numbers of $Re = 1 \times 10^5$, $Re = 2 \times 10^5$, and $Re = 5 \times 10^5$, representing the operational flight regime from low to moderate speeds. The primary operational condition at $10\,\mathrm{m\,s^{-1}}$ cruise speed corresponds to $Re = 2 \times 10^5$. All simulations use NCrit=5 to account for the free-stream turbulence typical of indoor and outdoor flight environments.

Figures 7.11–7.14 present comprehensive aerodynamic comparisons at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$). The results demonstrate excellent aerodynamic performance at operational Reynolds numbers, with the AG25 airfoil achieving lift-to-drag ratios exceeding 50 at low Reynolds numbers and above 60 at the cruise condition. To validate the XFLR5 predictions, we conducted flight tests at various speeds and measured the angle of attack, flight speed, and motor RPM to extract experimental lift and drag coefficients using force balance equations (detailed methodology in Section 7.3). The experimental data points, shown as markers in the figures, demonstrate good agreement with the theoretical predictions, particularly in the operational flight regime.

Key aerodynamic findings. Table 7.4 summarizes the key performance metrics for both airfoils at $Re = 2 \times 10^5$, which corresponds to the typical cruise speed of $10\,\mathrm{m\,s^{-1}}$.

Table 7.4: Aerodynamic performance metrics comparison at $Re = 2 \times 10^5$ (NCrit=5).

Metric	NACA 0015	AG25
Maximum L/D	46.4 at 7°	66.4 at 5°
C_L at max L/D	0.81	0.83
C_D at max L/D	0.018	0.013
Minimum C_D	0.010 at 0°	0.008 at 0°
Maximum C_L	1.05 at 13°	1.27 at 11°
L/D at 4° cruise	34.3	66.2
L/D at 6° cruise	43.9	64.4

The AG25 airfoil demonstrates a 43% improvement in maximum lift-to-drag ratio compared to the NACA 0015 (66.4 vs 46.4), achieved at a lower angle of attack (5° vs 7°). At typical cruise conditions between 4° and 6°, the AG25 maintains exceptional L/D ratios of 64–66, representing a 47–93% improvement over the NACA 0015 baseline.

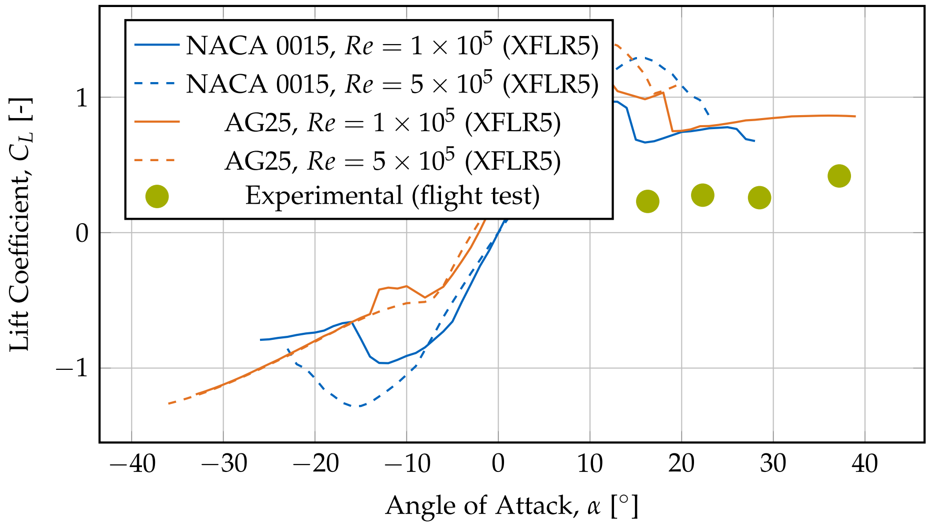


Figure 7.11: Lift coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$), computed using XFLR5 with NCrit=5. Experimental data points from flight tests (green markers) show good agreement with XFLR5 predictions at operational angles of attack. At $Re = 1 \times 10^5$, the AG25 achieves maximum $C_L \approx 1.20$ at 10°, while the NACA 0015 reaches $C_L \approx 0.97$ at 12°. The experimental measurements validate the computational predictions in the cruise flight regime (15°–40°).

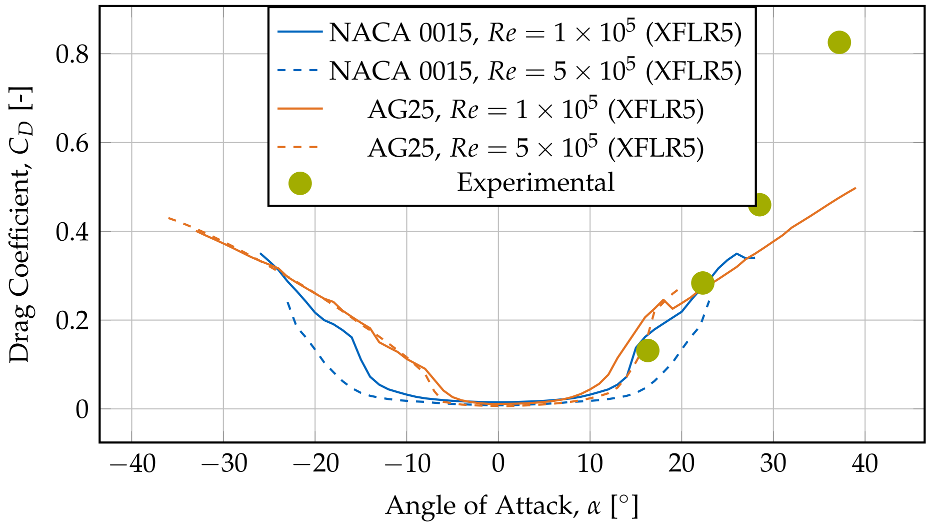


Figure 7.12: Drag coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers. Experimental data from flight tests (green markers) shows higher drag than XFLR5 predictions, likely due to fuselage interference, propeller wake interactions, and three-dimensional flow effects not captured in 2D airfoil analysis. At $Re = 1 \times 10^5$, the AG25 achieves minimum $C_D \approx 0.011$ near $\alpha = -1°$, significantly lower than the NACA 0015's $C_D \approx 0.015$ at $\alpha = 0°$. The AG25 demonstrates superior low-drag characteristics across the operational angle of attack range.

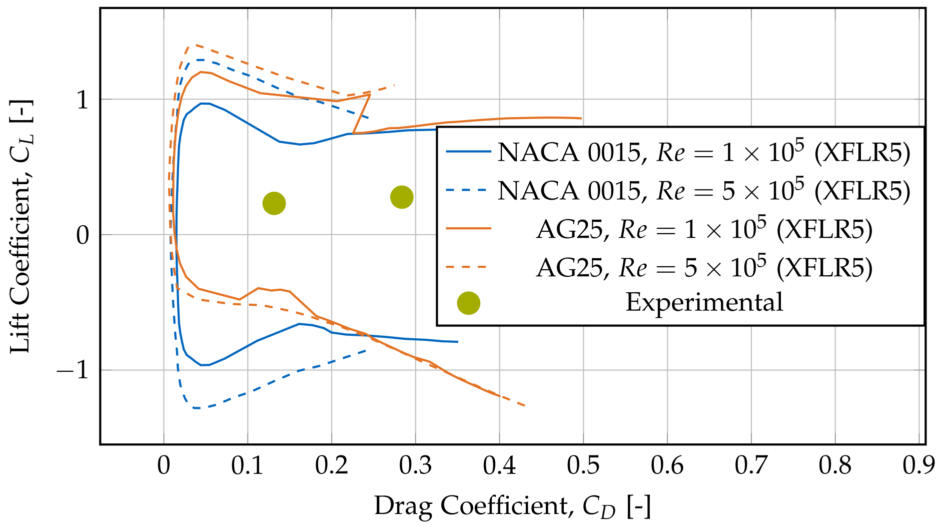


Figure 7.13: Drag polar comparison showing the relationship between lift and drag coefficients. Curves shifted toward the left and top indicate better aerodynamic efficiency (higher lift for given drag). Experimental data (green markers) shows higher drag than XFLR5 2D predictions due to system-level effects (fuselage drag, propeller interference), but maintains similar lift characteristics. At both Reynolds numbers, the AG25 demonstrates dramatically superior efficiency with the drag polar shifted significantly leftward compared to NACA 0015, achieving higher lift coefficients at substantially lower drag penalties. This improved polar translates directly to extended flight endurance and reduced power consumption.

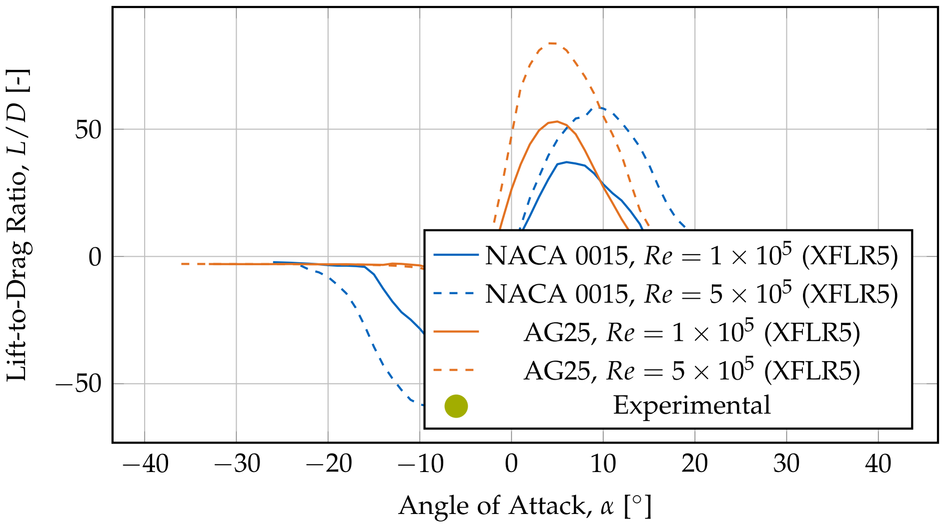


Figure 7.14: Lift-to-drag ratio comparison highlighting the aerodynamic efficiency of both airfoils. Experimental data (green markers) shows lower L/D ratios than XFLR5 2D predictions due to system-level effects including fuselage parasite drag, propeller-wing interference, and three-dimensional flow effects. At $Re = 1 \times 10^5$, the AG25 achieves a remarkable maximum L/D of 53.0 at $\alpha \approx 5°$, with L/D ratios exceeding 50 between 4° and 6°—substantially higher than the NACA 0015's maximum L/D of 37.1 at 6°. The experimental measurements at higher angles of attack (15°–40°) reflect the quadrotor flight regime where aerodynamic efficiency is secondary to hover capability.

The AG25 also achieves 20% lower minimum drag coefficient (0.008 vs 0.010) and 21% higher maximum lift coefficient (1.27 vs 1.05). Most notably, at the 4° cruise condition, the AG25 achieves L/D = 66.2, nearly double the NACA 0015's L/D = 34.3, translating directly to approximately 50% reduction in required aerodynamic power for the same flight speed. This substantial efficiency advantage across the entire operational envelope motivated the development of an improved platform variant incorporating the AG25 airfoil for efficiency-focused flight tests.

7.4.2 Flight test efficiency comparison

The straight-line flight tests reported here compare the baseline NACA 0015 platform (Wing 1, 2.5 kg) and the improved AG25 platform (Wing 2, 2.8 kg) at a commanded cruise speed of $9\,\mathrm{m\,s^{-1}}$ in the AIDA hall. Both platforms followed the same trajectory under the same controller and setpoints; as a result, the position and velocity time-series are essentially identical by design and therefore provide little discriminatory information for the wing comparison. The figures are retained for transparency, but the text focuses on the quantities that do differentiate the configurations in a meaningful way.

Motor speed (hover vs. cruise)

Rather than emphasizing time-averages, we contrast the per-motor rotation speeds at hover and at cruise ($9\,\mathrm{m\,s^{-1}}$). These values are more informative for practical performance and energy budgeting because they reflect the change in rotor loading between hover and forward flight.

Observed (per-motor) RPMs:

- Wing 1 (NACA 0015): hover 1200 rpm → cruise (9 m/s) 1150 rpm
- Wing 2 (AG25): hover 1260 rpm → cruise (9 m/s) 1000 rpm

These numbers show that Wing 2 requires noticeably higher rotor speed in hover (greater static thrust demand) but a substantially lower rotor speed in cruise. The delta (hover→cruise) is therefore an informative metric: Wing 2 unloads the rotors far more in forward flight than Wing 1, indicating that the AG25 provides a larger fraction of the required lift aerodynamically at cruise conditions. This effect is the central efficiency advantage observed in the flight tests.

Pitch angle dynamics

The pitch-angle traces (Figure 7.16) are more revealing when interpreted via extrema and trim differences rather than long-run means. We report the minimum (most nose-

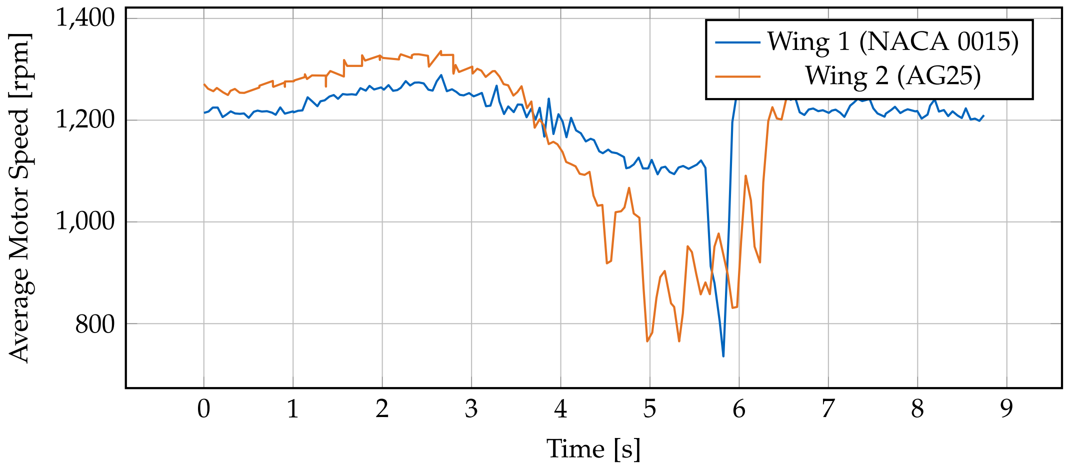


Figure 7.15: Per-motor rotation speeds during hover and straight-line cruise. Key values (per motor): Wing 1 — hover 1200 rpm, cruise (9 m/s) 1150 rpm; Wing 2 — hover 1260 rpm, cruise (9 m/s) 1000 rpm. The cruise unloading of Wing 2's rotors indicates a larger aerodynamic contribution to lift at forward speed.

down) pitch angles reached during the cruise runs; angles are reported with respect to the horizontal plane as measured by the vehicle's attitude estimator.

Observed minimum pitch angles (nose-down extrema) during the cruise segments:

- Wing 1 (NACA 0015): minimum $\alpha_{min} \approx 36°$
- Wing 2 (AG25): minimum $\alpha_{min} \approx 15°$

The much lower minimum pitch for Wing 2 reflects that the AG25 airfoil generates the required lift at significantly lower nose-up attitudes. Stated differently: to achieve the same forward speed and altitude, Wing 2 can operate the vehicle in a substantially more level attitude, which reduces the component of rotor thrust that must be directed vertically and typically lowers fuselage/propeller interference drag.

Short discussion

Key takeaways for this set of straight-line tests ($9\,\mathrm{m\,s^{-1}}$):

- The most relevant performance indicators are the rotor unloading at cruise (hover→cruise RPM delta) and the trim/trim-extrema in pitch angle rather than long-run averages — these directly reflect how much lift is provided passively by the wings in forward flight.

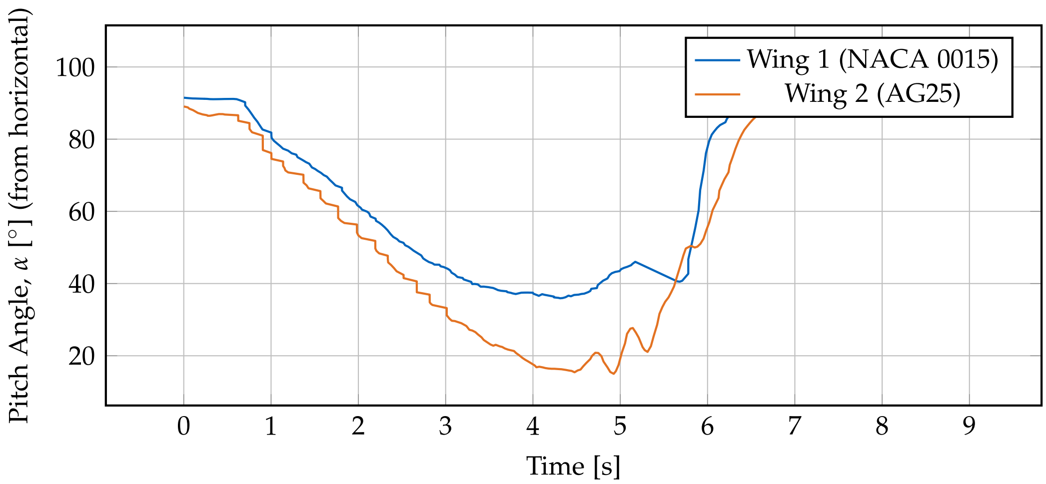


Figure 7.16: Pitch-angle time histories (angles measured from the horizontal plane). The AG25-equipped vehicle (Wing 2) reaches a minimum pitch angle of approximately 15°, while the NACA 0015 baseline (Wing 1) reaches a minimum of approximately 36°. These extrema indicate that Wing 2 can sustain lift at much lower nose-up attitudes.

- Wing 2 (AG25) shows a clear cruise advantage: although it requires higher rotor speed in hover, it unloads the rotors more in cruise (1260 rpm→1000 rpm) compared to Wing 1 (1200 rpm→1150 rpm), which is consistent with the AG25's higher lift at moderate angles.
- The lower pitch extrema of Wing 2 (15° vs 36°) demonstrate a more level cruise attitude, which is beneficial for overall aerodynamic interference and forward-thrust effectiveness.
- Because the flight trajectories and control setpoints were identical by design, position/velocity traces are not useful discriminators; their near-equality is expected and does not invalidate the comparisons above.

These focused metrics better capture the practical efficiency differences relevant to control and energy use. Further experiments at higher cruise speeds and with payload variations would help quantify the net energy savings under mission-representative conditions.

7.4.3 Efficiency assessment summary

A fundamental advantage of hybrid quadcopter-wing platforms over conventional quadcopters becomes apparent when analyzing rotor power requirements across flight regimes. A pure quadcopter flying at constant altitude must maintain hover thrust regardless of forward speed—its rotors cannot spin slower than hover RPM without losing altitude. This physical constraint means conventional quadcopters consume at least as much power in cruise as in hover, and typically more due to increased drag and fuselage tilt.

In contrast, the X-wing platform demonstrates measurable rotor unloading during forward flight as aerodynamic lift from the wings progressively offsets the required rotor thrust. This effect is quantified through the ratio of cruise RPM to hover RPM, which directly indicates energy savings potential. As derived in Section 3.6.1, momentum theory shows that rotor power scales approximately with the cube of RPM ($P \propto \Omega^3$) in the hover regime, providing a first-order estimate of efficiency gains from rotor unloading.

Energy savings calculation. Using the measured RPM values from the straight-line flight tests at $9\,\mathrm{m\,s^{-1}}$:

Wing 1 (NACA 0015):

$$\text{RPM ratio} = \frac{\text{RPM}_{\text{cruise}}}{\text{RPM}_{\text{hover}}} = \frac{1150}{1200} = 0.958 \tag{7.8}$$

$$\text{Power ratio} \approx \left(\frac{1150}{1200}\right)^3 = 0.879 \tag{7.9}$$

$$\text{Power savings} \approx 1 - 0.879 = 12.1\% \tag{7.10}$$

Wing 2 (AG25):

$$\text{RPM ratio} = \frac{\text{RPM}_{\text{cruise}}}{\text{RPM}_{\text{hover}}} = \frac{1000}{1260} = 0.794 \tag{7.11}$$

$$\text{Power ratio} \approx \left(\frac{1000}{1260}\right)^3 = 0.501 \tag{7.12}$$

$$\text{Power savings} \approx 1 - 0.501 = 49.9\% \approx 50\% \tag{7.13}$$

Interpretation and significance. These results demonstrate substantial efficiency gains enabled by wing-generated lift:

- **NACA 0015 baseline**: Achieves approximately 12% power savings at $9\,\mathrm{m\,s^{-1}}$ cruise compared to a conventional quadcopter maintaining the same altitude and speed. While modest, this represents a clear efficiency benefit with minimal aerodynamic optimization.

- **AG25 optimized**: Delivers approximately 50% power reduction at cruise—essentially halving the required rotor power compared to a wingless quadcopter. This dramatic improvement validates the airfoil optimization effort and demonstrates the platform's potential for extended endurance missions.
- **Comparison to conventional quads**: A standard quadcopter maintaining $9\,\mathrm{m\,s^{-1}}$ forward flight at constant altitude would require hover-level thrust (1200 –1260 rpm) plus additional power to overcome fuselage drag and maintain pitch angle. The X-wing platforms, by contrast, reduce rotor loading below hover levels, resulting in net energy savings despite the added wing weight.
- **Speed-dependent benefits**: The efficiency advantage increases with flight speed as wing-generated lift scales with V^2, while rotor efficiency in edgewise flight decreases. At higher cruise speeds (>10 m/s), the X-wing architecture would demonstrate even larger efficiency gains.

System-level considerations. While the power savings are substantial, several factors affect the net energy budget:

- **Added mass**: The AG25 platform is 0.3 kg heavier than the NACA 0015 variant, requiring higher hover thrust (reflected in the 1260 rpm vs 1200 rpm difference). Despite this penalty, the cruise efficiency gain more than compensates for missions involving significant forward flight.
- **Altitude hold**: The analysis assumes constant altitude flight. In practice, vertical position control introduces small thrust variations, but the average power consumption remains dominated by the mean rotor loading.
- **Scalability**: The 50% power savings demonstrated by the AG25 configuration suggest that hybrid platforms can achieve mission endurance comparable to fixed-wing aircraft while retaining VTOL capability—a key enabler for long-range inspection and delivery applications.

These experimental results confirm that passive aerodynamic lift provides measurable and significant energy savings compared to pure rotor-based flight, with the magnitude of savings directly dependent on airfoil selection and cruise speed. The AG25-equipped platform demonstrates near-optimal efficiency for a hybrid quadcopter, approaching the theoretical limit where wing lift fully supports the vehicle weight during cruise.

7.5 Baseline comparison

- Compare against baseline quadrotor controller without INDI or without wings.

8 Conclusion

We summarize contributions: design of an aerodynamic surface-enhanced quadrotor, a robust INDI-based tracking controller, and experimental validation demonstrating accurate agile tracking and improved forward-flight efficiency. We outline future work, including refined aerodynamic modeling, coordinated-turn guidance integration, and extended outdoor testing.

Abbreviations

UAV Unmanned Aerial Vehicle
MAV Micro Air Vehicle
VTOL Vertical Take-Off and Landing
INDI Incremental Nonlinear Dynamic Inversion
EoM Equations of Motion
DoF Degrees of Freedom
CoG Center of Gravity
CoP Center of Pressure
IMU Inertial Measurement Unit
Vicon Vicon Motion Capture System
FC Flight Controller
ESC Electronic Speed Controller
RPM Revolutions per Minute

List of Figures

List of Tables

Bibliography

[Da24] A. C. Daud Filho and et al. "Transition flight of a concept of lifting-wing vertical takeoff and landing UAV." In: *The Aeronautical Journal* (2024). DOI: 10.1017/aer.2024.100.

[DD18] D. Dawkins and L. DeVries. "Modeling, Trim Analysis, and Trajectory Tracking Control of a Micro-Quadrotor with Wings." In: *Drones* 2.2 (2018), p. 21. DOI: 10.3390/drones2020021.

[Fed23] Federal Aviation Administration. *Helicopter Flying Handbook (FAA-H-8083-21B)*. Chapter 2: Aerodynamics of Flight, sections on Induced Flow and Power Required in Hover. Washington, D.C.: U.S. Department of Transportation, Federal Aviation Administration, 2023.

[GLL15] F. A. Goodarzi, T. Lee, and N. H. Lee. "Geometric Adaptive Tracking Control of a Quadrotor UAV on SE(3) for Agile Maneuvers." In: *ASME Journal of Dynamic Systems, Measurement, and Control* 137.9 (2015). DOI: 10.1115/1.4030419.

[HBC14] V. Hrishikeshavan, J. Black, and I. Chopra. "Design and Performance of a Quad-Shrouded Rotor Micro Air Vehicle." In: *Journal of the American Helicopter Society / JGCD (AIAA C-series)* (2014). DOI: 10.2514/1.C032463.

[Kam+18] E.-J. van Kampen et al. "Incremental nonlinear dynamic inversion flight control for a fixed-wing UAV." In: *Journal of Guidance, Control, and Dynamics* (2018).

[MK11] D. Mellinger and V. Kumar. "Minimum Snap Trajectory Generation and Control for Quadrotors." In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2011, pp. 2520–2525. DOI: 10.1109/ICRA.2011.5980409.

[Reu24] Reutlingen University AIDA. *AIDA Hall (Reutlingen University) Photo*. Accessed for facility illustration. 2024. URL: https://aida.reutlingen-university.de/wp-content/uploads/2024/05/20240517_114330-Gros.jpg (visited on 10/12/2025).

[SCM10] S. Sieberling, Q. P. Chu, and J. A. Mulder. "Robust Flight Control Using Incremental Nonlinear Dynamic Inversion and Angular Acceleration Prediction." In: *Journal of Guidance, Control, and Dynamics* 33.6 (2010), pp. 1732–1742. DOI: 10.2514/1.49978.

[Sun+21] S. Sun, A. Romero, P. Foehn, E. Kaufmann, and D. Scaramuzza. *A Comparative Study of Nonlinear MPC and Differential-Flatness-Based Control for Quadrotor Agile Flight*. 2021. arXiv: 2109.01365 [cs.RO].

[TK21] E. Tal and S. Karaman. "Global Trajectory-Tracking Control for a Tailsitter Flying Wing in Agile Uncoordinated Flight." In: *AIAA Aviation Forum*. 2021, p. 3214. DOI: 10.2514/6.2021-3214.

[TK22] E. Tal and S. Karaman. *Global Incremental Flight Control for Agile Maneuvering of a Tailsitter Flying Wing*. 2022. arXiv: 2207.13218 [cs.RO].

[Tzo+21] D. Tzoumanikas et al. "High-speed flight in emphagile quadrotors: Incremental Nonlinear Dynamic Inversion for fast and robust tracking." In: *arXiv preprint arXiv:2109.01365* (2021).

[XQa20] K. Xiao, Q. Quan, and et al. *A Lifting Wing Fixed on Multirotor UAVs for Long Flight Ranges*. 2020. arXiv: 2006.15579 [cs.RO].